Continuous Scene Meshing for Dynamic Spatial Awareness in Mixed Reality

A Comparative Analysis of On-Device vs. Hybrid Edge Architectures for Real-Time Dynamic Environment TSDF Construction in Mobile Mixed Reality

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Abstract

Introduction Mixed reality applications increasingly need a three-dimensional model of the room that keeps updating so that virtual objects can collide with real walls and be hidden behind real furniture as that furniture moves. Building such a model continuously is computationally expensive, and mobile headsets lack the power to do this for extended hours. This has led the field toward architectures that offload the work to edge servers, on the assumption that doing so relieves the headset of a load it cannot sustain for long.

Research Question This thesis asks what the sustained thermal, energy, and frame-time behavior of continuous on-device reconstruction actually is, and whether the headset reaches thermal throttling within a realistic session; what bandwidth and latency offloading costs are, and how that cost divides between moving data and computing remotely; and under what conditions, if any, offloading beats on-device processing. It also asks whether users can perceive any difference between the two architectures, and whether that depends on the kind of interaction and on the quality of the network link.

Method Following Design Science Research (DSR), two functionally identical artifacts were built on a Meta Quest 3. One performs Truncated Signed Distance Function (TSDF) integration and Naive Surface Nets meshing entirely on the headset. The other streams depth data to an edge server running on a MacBook Pro over local Wi-Fi and applies the returned mesh. Both share the same reconstruction mathematics, so the main difference is where the work runs. Three instrumented runs of 18 to 41 minutes were recorded across two rooms, with the two headsets stacked and carried together. Hence, both architectures saw the same geometry at the same moment, and the architectures swapped between the physical headsets to separate architectural effects from hardware differences. Nine participants

then compared the two architectures across two interaction tasks and three degraded network conditions in two waves running mirrored task orders.

Results Neither architecture reached thermal throttling. On-device reconstruction settled at approximately $49\,^{\circ}\mathrm{C}$ after ten minutes and held that temperature for a further thirty. Offloading required about 6 Mbit/s upstream and 27–29 Mbit/s downstream, with roughly 95% of the round-trip delay spent moving data rather than computing remotely. Once the physical headset was controlled for, offloading consumed about 0.5 W more power than computing locally and produced far worse worst-case frame times. In a second room with weaker wireless conditions, it fell below one mesh update per second while the on-device path was unaffected. Participants distinguished the two architectures on only one task (the one requiring the reconstruction to be correct about geometry that had just moved) where eight of nine rated the offloaded architecture higher and the ninth rated them equal, in both presentation orders.

Discussion Offloading did not deliver the benefit that motivates it. Because the uncompressed mesh returning to the headset is larger than the depth stream that produced it, the radio cost more power than moving the computation saved, and no amount of server capacity helps. The ratio between an offloaded task's output and its input therefore appears to predict viability better than its computational cost does. The resulting dependency also failed unpredictably between ordinary indoor rooms, which for an all-day wearable device may matter more than the energy question. The conditions under which offloading wins are narrow but real: a local server must be available, the link must not be degraded, and the interaction must value a complete reconstruction above worst-case frame timing and above battery life.

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Finally, my thanks go to the participants who volunteered their time, moved around a room wearing a headset, and answered a great many questions about how it felt.

Abbreviations

Abbreviation	Description
API	Application Programming Interface
AR	Augmented Reality
CPU	Central Processing Unit
DSR	Design Science Research
DSRM	Design Science Research Methodology
GPU	Graphics Processing Unit
HLSL	High-Level Shading Language
HMD	Head-Mounted Display
MEC	Mobile Edge Computing
MR	Mixed Reality
NASA-TLX	NASA Task Load Index
RTT	Round-Trip Time
SLAM	Simultaneous Localization and Mapping
SoC	System on a Chip
TCP	Transmission Control Protocol
TSDF	Truncated Signed Distance Function
VR	Virtual Reality
VRSQ	Virtual Reality Sickness Questionnaire
XR	Extended Reality

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1

Introduction

Mixed Reality (MR) is moving from static, pre-configured environments toward models of the world that keep updating as the world changes. Over the past decade, the "scan and lock" paradigm has been the main choice for consumer MR devices, where the user scans their environment, and the system retains a static collision volume from this initial scan [4]. While this is appropriate for static environments, it is inadequate in the face of changing or active environments, such as social spaces, interactive performances, or collaborative workspaces, where objects and people move around. For example, to accurately simulate the interaction between a virtual ball bouncing off a moving person, the system constantly needs to update its understanding of the physical world. It must constantly sense its environment and have an up-to-date volumetric representation of the objects in the scene, at a fidelity and timescale that allows it to meaningfully interact with the physical world [9][16].

This requires a continuous reconstruction of the geometry within the simulated volume based on depth information of the physical world. While this is certainly possible, it is computationally intensive. A common approach to this problem is volumetric fusion, which uses Truncated Signed Distance Functions (TSDF) to represent the geometry. This approach was popularized by the KinectFusion system [9][16]. This method fuses depth images together to form a closed, watertight mesh. It is particularly good at representing surfaces, which is useful for physics simulations, as well as occlusion culling.

The problem with this approach is that, while it produces great results, it is extremely computationally intensive and requires a large amount of memory bandwidth. At 30-60 fps, a fine voxel grid could require far more processing power than a standalone battery-powered headset can provide [10].

1.1 Problem

The main issue for newer all-day XR wearables is computational load. People want devices that feel like glasses, not like bulky headgear. That choice cuts down the space where heat can dissipate. Heat transfer rules make this a strict constraint. A system that stays on and watches everything cannot run at its full power without either slowing down due to overheating or draining the battery too fast [5] [10] [18]. As a result, architectures are moving toward designs that use the edge servers more. They send the heavy reconstruction work to edge servers through fast wireless links [14].

Even with this theoretical promise, the practical trade-offs are unknown and not researched enough for modern hardware. More specifically, no studies have mapped the thresholds at which the energy saved by offloading outweighs the added cost of network transfers and delays in systems that keep meshing all the time.

Whether offloading actually pays off is an open question. For it to be worth it, the energy saved by skipping local work must be *more* than the energy spent sending the depth data and receiving the result. It must also cover the energy tied to waiting for and getting the reconstructed output. In dense volumetric reconstruction, this output can be large. This study carried out a direct comparison of two different software architectures for continuous spatial mapping to examine this trade-off.

1. **On-Device Architecture:** A fully autonomous system where depth processing, volumetric fusion, and mesh extraction occur locally on the mobile headset's System-on-Chip (SoC).

2. **Hybrid Remote Edge Architecture:** A distributed system where raw depth data is streamed to a local edge server for processing, with the resulting physics mesh streamed back to the headset.

In computer and systems science, this study contributes to mobile edge computing and human–computer interaction (HCI) domains. It brings the two areas together in a single investigation. The object of study is an architectural decision, where the perception workload executes, which is a

systems question, so the metrics are in watts, milliseconds, and megabits per second. In addition to the system-level metrics, the evaluation considers the end-user experience. Specifically, it investigates whether the user can notice a difference between two system setups. The users in this scenario are not aware of the variation. Prior studies often considered the system and user-level considerations as two separate aspects. An important insight of this thesis is that these two views are inseparable for tasks involving continuous reconstruction. RQ4 is where that second area enters the design of the study: it is answered by asking participants rather than by instrumenting the device.

1.2 Research questions

The thesis investigates, through an implementation and experiments, if and under what conditions offloading the task of continuous dense reconstruction to an edge server is viable for mobile mixed reality. **It does not pose the question of finding an optimal system but rather addresses the problem as an optimization space exploration.** The question is motivated by observations from the literature claiming that offloading is beneficial for thermally constrained headsets. However, those works do not evaluate the hypothesis on real devices. This thesis attempts to address this limitation by performing such an evaluation.

The work uses the following artifacts to compare the viability of the two approaches with regard to real-time dynamic TSDF reconstruction and answer the research questions presented below:

- **RQ1: On-device limits.** What are the sustained thermal, energy, and frame-time characteristics of a continuous meshing system running entirely on a standalone mobile headset? Is thermal throttling reached within a realistic session length?
- **RQ2: The cost of offloading.** What bandwidth and latency does offloading dense reconstruction to a local edge server actually cost, and how does that cost divide between network transport and remote computation?
- **RQ3: Conditions for viability.** Under what conditions, if any, does the offloading architecture outperform on-device processing in energy use and frame-time stability?

- **RQ4: Perceived difference.** Can users perceive a difference between the two architectures, and does that difference depend on the kind of interaction being performed and on the quality of the network link?

1.3 Delimitations

The hardware setup was restricted to Meta Quest 3 headset for the experiments. As for the edge server, MacBook Pro with an Apple M3 Max chip was used. All the tests were conducted in the same local Wi-Fi network. Because the study is about offloading to a machine in the same building, cellular links, wide-area latency and long-distance propagation are out of scope. Network degradation is not: it is induced deliberately under controlled conditions in the user study (Chapter 3).

The evaluation process comprises two studies which investigate different spatial contexts. While the instrumented study considers indoor spaces simulating a domestic and an office environment, it includes variations in terms of room size and shape complexity. Instead of referring to area size, the unit of measurement considered for the complexity of the space is the mean vertex count of the reconstructed mesh resulting from the experimental process. This choice reflects the impact that a higher number of vertices has on the computational demand of the reconstruction process.

The user study took place across four different indoor rooms, with each participant using one room for their whole session. Participants stood within roughly 1×1 meter of clear floor space and turned on the spot, since the targets orbit the player rather than requiring the player to cross the room. The rooms were not held constant, so the environment is not controlled in the user study the way it is in the instrumented study. What was held constant is the distance to the wireless access point, which stayed within 1–3 meters in every room, the same range as the cluttered room in the instrumented study, where the offloaded architecture performed normally. Because the comparison between architectures is made within each participant's session, differences between rooms cannot explain it.

1.4 Thesis outline

Chapter 2 provides the background technical knowledge for the study. In particular, it presents volumetric fusion, the thermal envelope of mobile systems-on-chip, and the structure of the two artifacts being evaluated.

Furthermore, it also places the work in the context of previous research on edge-assisted augmented reality and energy profiling.

Chapter 3 describes the Design Science research methodology applied in the study. In addition, it details the two evaluation studies, including their instrumentation and procedures.

Chapter 4 details the functional and non-functional requirements for a solution to the continuous reconstruction problem. Furthermore, describes how the requirements are connected to the research questions.

Chapter 5 presents the results of the evaluation. Specifically, it includes functional verification of both artifacts, performance measurements from the instrumented study, and user study results.

Chapter 6 discusses and interprets the results. In addition, it considers the study's limitations and outlines directions for future work.

2

Extended Background

This chapter presents the study's theoretical and technical concepts, as well as the environment in which the research was conducted. It also describes what the examined artifacts do and how they are constructed, and presents relevant prior academic research to identify gaps in the field.

2.1 The Evolution of Spatial Mapping

Early Simultaneous Localization and Mapping (SLAM) systems focused on using sparse feature points to track device position by matching these points in the surrounding scene [12]. While these methods could position virtual objects roughly, their inability to represent surface structure accurately enough made them unsuitable for capturing physical characteristics. For example, sparse mapping could not recreate an image of a ball hitting a wall and bouncing off.

Later iterations of SLAM, however, focused on dense reconstruction of the scene itself rather than only the camera's path through it. One example is the "Scene Mesh" function in the Meta Quest 3, which uses depth sensing to generate a polygon mesh of the tracked volume [6]. However, because mesh tracking often does not update as quickly as the visuals, applications that use depth information often simulate a static scene while the camera moves. To track depth information in real time, an additional process of continuous depth fusion must take place [9][16].

2.2 Truncated Signed Distance Functions (TSDF)

The industry's standard approach to merging noisy depth frames into a 3D model is the TSDF process, which has seen a resurgence through KinectFusion [9][16]. The scene is represented as a voxel grid, where each voxel stores a value corresponding to distance from the nearest surface. As the camera moves, subsequent depth frames are integrated into the grid to reduce noise and fill in missing data. An isosurface extraction algorithm (Marching Cubes classically, and Naive Surface Nets in this work) then converts the voxel grid into a renderable mesh. This approach is computationally intensive and requires significant memory, scaling cubically with volume resolution [9][13][16], and is therefore a good candidate for offloading.

2.3 Thermal Constraints

Mobile SoCs such as the Snapdragon XR2 Gen 2 in the Quest 3 are capable of processing demanding applications. However, they must also remain within strict thermal limits, and meshing as part of a continuous process is a particularly long-tail workload that taxes the CPU with concurrent application processing. The GPU must perform both rendering and volumetric integration steps. The digital signal processor is also used heavily as it is involved in decoding depth information [15].

As the temperature rises toward the level at which the device is rated to throttle, the system will have to reduce clock speeds to dissipate the heat, ultimately governed by a safety threshold that relates back to human skin tolerances. Reducing clock speed causes dropped frames, which lengthen the motion-to-photon delay and can induce simulator sickness.

Whether such a workload will hit the worst-case scenario during a typical session is an open question and is the subject of this thesis.

2.4 Practice and Problem Context

Within the practice of mobile Mixed Reality, the defining limitation is no longer raw rendering capability but thermal dissipation and energy sustainability [18]. The industry has largely abandoned tethered PC-VR setups in favor of standalone, battery-powered Head Mounted Displays such as the Meta Quest 3. These devices are expected to run complex perceptual algorithms with Simultaneous Localization and Mapping, hand tracking,

and spatial meshing, concurrently with high-resolution stereoscopic rendering.

Both evaluated artifacts address the same fundamental problem: *the computational bottleneck of continuous dense surface reconstruction.* Unlike sparse SLAM, which tracks only a set of feature points to determine camera pose, dense reconstruction attempts to model every physical surface in the environment. The standard approach is TSDF fusion, a highly parallelizable but memory-bandwidth-intensive volumetric algorithm [9].

Mobile SoCs operate within a very small thermal envelope, so sustained compute is bounded by heat dissipation rather than by peak throughput. A recent feasibility study of ARM-based SoCs on the Quest 3 reports that continuous real-time depth processing leaves very little thermal headroom on the XR2 Gen 2 [15]. When the governor throttles CPU and GPU clocks in response, the resulting dropped frames threaten the motion-to-photon budget that comfortable mixed reality depends on [18].

The idea of offloading also drives this thesis, and rather than taking it as a given, it tests it directly. From that test, two hypotheses emerge. These are: **a standalone on-device meshing will hit thermal throttling during a normal session, and running the work on an edge server lowers the headset's energy use.** Chapter 5 checks both with experiments.

2.5 Artifacts

To test these trade-offs, two software prototypes were built. Both run the same TSDF reconstruction algorithm, but they differ fundamentally in structure, behavior, and where they run.

2.5.1 Architecture A: On Device (Baseline)

The artifact consists of a Unity application based on a fork of the *LaserTag* project repository [1]. The depth information is provided by the Unity AR Foundation [2]. Specifically, the occlusion system is implemented using the Meta OpenXR Plugin, which uses the Quest 3 depth sensor [6]. Instead of manually calculating the projection and view matrices for each frame, the application subscribes to the occlusion manager's frame event. Within the event, the stereo depth texture, per-eye field of view, pose, and projection matrix (including near and far plane values) are available for use. The event handler uses this data to calculate the projection and view matrices for reconstruction. After processing the event, the application passes the

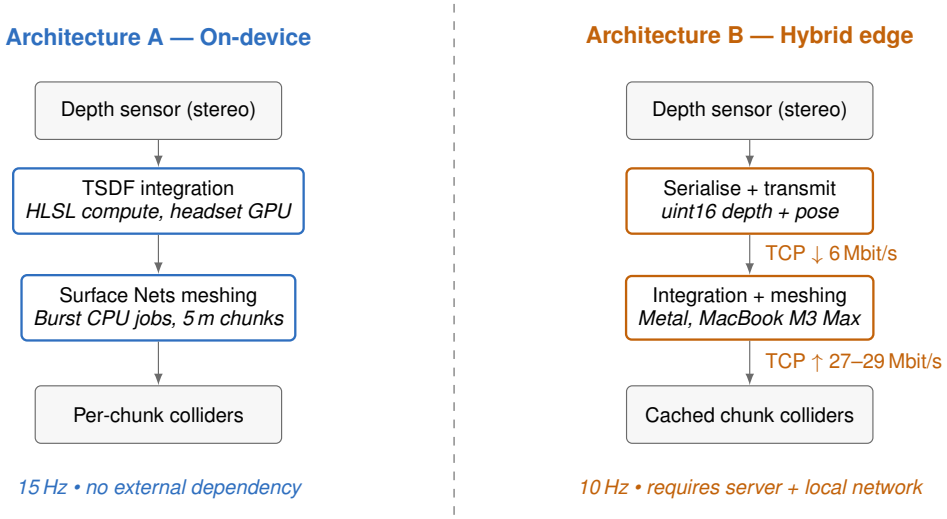


Figure 2.1: The two reconstruction pipelines. Both implement identical TSDF mathematics and the same Naive Surface Nets algorithm; they differ only in where integration and meshing execute. The lower offloaded update rate is a consequence of that (Section 3.2.1). Grey stages are shared by both architectures.



Figure 2.2: The reconstruction as it is built, captured from the headset during an instrumented run. The mesh is shown here to illustrate the output of the pipeline; **it was visible only in the instrumented builds and was hidden in the user-study build**, so that participants judged the interaction rather than the appearance of the mesh.

calculation to the headset, which uses a custom HLSL compute shader to perform it.

The application ingests the stereo depth frames and combines them into a single TSDF volume in a spatial hash grid. The grid update frequency is set to 15 Hz. During processing, the compute shader converts depth values into voxel samples and updates signed distance values to form a 3-D volume. The resulting volume is then used to generate a set of meshes by running Naive Surface Nets algorithms, implemented in Burst as part of the CPU job. The application's final output is a set of per-chunk meshes used as physics colliders.

In the game the player holds a virtual blaster and fires bullets at floating virtual targets. The bullets and targets are virtual, the room is not. Bullets are where the reconstruction becomes visible, because they stop against real walls and furniture.

The design goal is low sensing latency rather than low power. Take a real door.If the door opens, Architecture A refreshes the voxel grid right away. Then the bullets go through the opening without much delay. This speed comes with a cost. Each step can hit the headset's own thermal and power limits.

2.5.2 Architecture B: Hybrid Edge (Offloaded)

This artifact decouples the workload between a client and a server, rather than performing all tasks on the headset. The headset becomes a simple client, and a MacBook Pro (Apple M3 Max) on the same network hosts the edge server, a Swift application built on Metal compute shaders.

The headset no longer performs all depth-processing computations. Rather, it packages up the raw depth buffers and camera pose matrices and sends them over a local Wi-Fi connection via a plain TCP socket (10 Hz, one request in flight at a time).

The server application consumes the stream, incorporates the data into the server's TSDF volume, and performs geometry reconstruction using the same Naive Surface Nets algorithm used on the device. The Metal code directly ports the previous HLSL compute shader. The server then serializes the resulting mesh pieces and streams them back to the headset for caching. Unity application consumes the resulting visual primitives and applies them to the physics engine.

The idea with this architecture is that by offloading the intensive computation, the headset has more thermal capacity and can keep a session alive for longer. When the door opens, the headset streams the depth data,

the server builds the geometry, then streams the resulting mesh back to the headset.

With a stable network connection, the virtual bullets behave the same as in Architecture A. But as the connection degrades, bullets may hit the now-open space where the door was for a few seconds before the meshing updates the volume back, since the old mesh is not replaced until the next server update, resulting in poor interactive quality. This phenomenon is discussed further in [14].

Figure 2.1 shows both pipelines side by side. Because both use the same reconstruction algorithm, any differences in results must be attributed to differences between the systems.

2.6 Related Work

2.6.1 Edge-Assisted Augmented Reality

Offloading computation to a nearby server is well studied. AccuMO [14] presents a multi-task offloading framework for mobile AR and shows that offloading everything is a poor strategy, proposing dynamic scheduling instead. Edge-SLAM [3] splits the SLAM pipeline, keeping latency-critical tracking on the device and sending heavy map optimization to the server.

Two limits matter here. First, in both cases what comes back is a *small* camera pose, or correction to a sparse map. The question of what happens when the reply is bigger than the request simply never comes up for sparse tracking. Second, both are judged on system metrics alone, with no measurement of whether anyone using the system notices. The question they raise is the right one. But their answer is untested against dense reconstruction and untested against people.

2.6.2 Energy Profiling in Virtual Reality

Any claimed offloading benefit needs proper energy measurement. Kattakinda et al. analyze energy use in VR applications and identify GPU utilization as the main reason for battery drain [10]. Their method (deriving instantaneous power from sampled voltage and current rather than from battery percentage) is adopted as the measurement standard here, implemented through a purpose-built logger rather than the OVR Metrics Tool so that power, temperature, frame timing and network data land in one time-aligned record.

Wang et al. establish that offloading cuts local computation energy but raises transmission energy through radio power [20]. A break-even point therefore exists where sending the data costs more than processing it locally.

Neither settles the question for this workload. Wang et al. show the break-even point must exist, but where it falls depends on how much data is involved [20], so it has to be measured case by case. Kattakinda et al. provide the measurement technique but use it within a single device, where nothing crosses a network. Putting the two together (their technique, applied across a client–server boundary, on a workload whose reply is bigger than its request) is what this thesis adds.

2.6.3 Measuring the Human Response

Because this study compares architectures partly on how they feel, it needs instruments for perception as well as for power. Each carries a limitation that shapes how its results can be read.

Workload is measured with the NASA Task Load Index [7], a six-subscale instrument long established in human factors and widely used in XR research, applied here raw and unweighted. One property matters: its Performance subscale runs backward, from perfect to failure, while every other scale in this study runs from low to high. That inversion is a known source of respondent error, and Chapter 5 reports that it occurred.

Simulator sickness is measured with the Virtual Reality Sickness Questionnaire [11], nine items derived from the older Simulator Sickness Questionnaire and shortened for head-mounted displays. Its symptoms accumulate over time, so at the round lengths used here it is working in the region where it is least likely to detect anything.

Perceived reconstruction behavior has no validated instrument at all. No existing questionnaire asks whether a live reconstruction was correct about geometry that had just moved. Custom Likert items were written for this study and are documented in Chapter 3. They separate two ideas used throughout the later chapters: *freshness*, how quickly a change appears in the mesh, and *completeness*, how much of the surface is represented at all. An architecture can be better at one and worse at the other.

2.6.4 Gaps in How These Systems Are Evaluated

Prior Mobile Edge Computing work has explored the algorithms thoroughly but evaluates them narrowly [17]. Two patterns recur. Comparisons lean

on simulation rather than controlled experiments on physical hardware, which cannot reproduce the thermal behavior of a specific chipset. Analyses are almost entirely quantitative, relying on network logs while neglecting questionnaires or interviews that would capture human perception.

For MR this is a serious flaw. **A recorded 50-millisecond delay does not matter if nobody can see it, and a small delay may still break an interaction if it lands at the moment the world changes**.

2.7 Summary and Research Gap

The literature points to a consistent tension. Continuous dense reconstruction is what lets a mixed-reality system interact correctly with a changing world, and TSDF fusion is the established way to produce it. But it is expensive, and mobile headsets are limited by heat rather than by peak throughput. The field has settled on offloading to an edge server, supported by evidence that partitioned offloading beats wholesale offloading for sparse tracking and by the existence of an energy break-even point.

What the literature does not establish is whether that response works for *this* workload. The offloading results come from sparse tracking, where the reply is small. A dense reconstruction sends back a mesh, and no prior study measures the cost of that reversal. The break-even point is described in theory but never studied on consumer hardware. Evaluations lean on simulation, which cannot reproduce the thermal behavior that motivates offloading in the first place. And they report system metrics almost exclusively, leaving open whether any measured difference is perceptible to the person wearing the device.

This thesis addresses that gap by building both architectures from one codebase, measuring them simultaneously on real hardware, and pairing those measurements with a user study built around an interaction that only a correct and current reconstruction can support.

3

Methodology

This chapter sets out the approach, working down from the overall research strategy to the specific methods for collecting and analyzing data, so that another researcher could repeat the process and check the findings.

3.1 Method Choice

3.1.1 Research Paradigm and Framework

This thesis is within the Design Science Research (DSR) paradigm, it evaluates and compares two built IT artifacts (*Architecture A* and *Architecture B*) that address a technical problem arising from hardware limits in mobile mixed reality [8]. It follows the Design Science Research Methodology (DSRM) framework of Peffers et al. [19]. Design and development are covered by building the two prototypes.

Consistent with DSR's iterative character, *the research questions were refined during evaluation.* The original wording assumed throttling would occur within a session and that a crossover point existed beyond which offloading became worse than on-device processing. Neither assumption survived measurement, so the questions were reformulated (Section 1.2). This is reported here rather than presented as the original design.

3.1.2 Research Strategies

The study combines two strategies [17]. The **instrumented study** is a controlled experiment addressing RQ1 to RQ3. It varies the processing location and the room while measuring energy, temperature, frame time, and network behavior, following controlled energy profiling practice in mobile computing [10]. The **user study** is a survey addressing RQ4. It exists because a measured latency does not map directly onto what a person notices: a delay that shows up clearly in the telemetry may be invisible in use, and a delay that measures as small may still break an interaction if it lands at the moment the room changes. Instrumentation cannot answer that question, so the study asks nine participants to rate the two architectures directly.

3.1.3 Alternatives Considered

Two alternatives were rejected. **Software simulation**, common in edge computing research, cannot reproduce the thermal behavior, battery discharge, and cooling limits of the Snapdragon XR2 Gen 2, and those physical limits are the object of study, so testing on the device was mandatory. A **naturalistic field study** on a public network was also rejected. Unpredictable interference would make it impossible to attribute a frame-rate drop to the architecture rather than to a random network event. The cost of that choice is ecological validity, and the findings should be read with it in mind.

3.2 Artifacts and Apparatus

The client is a single Unity 6 application deployed to a Meta Quest 3. One project produces both architectures. A compile-time define selects the mode, and the two builds carry different Android package identifiers so both can be installed at once. A third define selects the user-study configuration, in which virtual targets are spawned, and the mesh is hidden from the participant. The mesh remained visible in the instrumented builds, where no participant was present and it was hidden only in the user-study build, so that participants judge the interaction rather than the appearance of the reconstruction. The display ran at 72 Hz throughout, recorded per run in the telemetry.

Depth is acquired identically in both architectures, through AR Foundation 6.4's occlusion manager [2], backed on this hardware by Meta's OpenXR

provider and the Quest 3 depth sensor [6]. The client subscribes to the occlusion manager's frame event rather than polling. Each frame supplies a stereo depth texture with the per-eye field of view, pose, and near/far planes needed to project it. Because acquisition is shared, it cannot contribute to any measured differences.

In both architectures, the depth frame is then dilated to fill holes, converted to normals, integrated into a TSDF volume of $1024 \times 256 \times 1024$ voxels at 0.1 m resolution (a physical extent of $102.4 \times 25.6 \times 102.4$ m), and meshed using Naive Surface Nets. Integration is truncated at 0.2 m and restricted to depth between 1 and 6 m.

- **Architecture A (On-device).** Integration runs in an HLSL compute shader on the headset GPU at 15 Hz; meshing runs on the CPU in Burst-compiled jobs over 5 m chunks. No data leaves the device.
- **Architecture B (Hybrid edge).** The client serializes raw depth buffers and camera matrices and streams them over a raw TCP socket to a MacBook Pro (Apple M3 Max) on the same Wi-Fi network. A Swift server integrates and meshes them with Metal compute shaders and returns 3.2 m triangle-mesh chunks, which the client caches and applies as colliders. Depth is sent as unsigned 16-bit values at 10 Hz with one request in flight at a time.

Both devices used a single Asus RT-AX55 access point on the 5 GHz band under Wi-Fi 6, with no other traffic deliberately introduced. The distance from headset to access point was not held constant across rooms; Section 3.3 records it, and it matters more to the results than any property of the server.

3.2.1 Why the two update rates differ

Both architectures run the same 15 Hz loop. On-device, each tick integrates the current depth frame. Offloaded, local integration is skipped, and the tick offers the frame to the network path, which adds a 100 ms minimum send interval and a rule that only one request may be in flight. The second is binding: the client will not send again until the previous reply arrives, so the send rate cannot exceed the reciprocal of the round-trip time.

The rate gap is a consequence of offloading, not a badly chosen parameter. A remote pipeline cannot run at 15 Hz across a 105 ms link with one request outstanding, and allowing concurrent requests would trade that bound for unbounded queuing and out-of-order mesh data. At the

651 ms median measured in the hallway, the rule permits at most about 1.5 updates per second, whatever the bandwidth. Otherwise both paths are identical, so the manipulated variable is where execution happens.

3.3 Data Collection

3.3.1 Instrumented Study

Two Quest 3 headsets were stacked one on top of the other and carried together, so both architectures saw the same room, motion, and lighting at the same moment. Running both at once rather than sequentially removes the largest sources of variance like walk path, pace, room state, and lighting. These are shared by construction rather than reproduced by hand. The principle follows the controlled sensor motion of the original KinectFusion evaluation [16], by simpler means.

The headsets were not rigidly fixed together, leaving a small vertical offset and allowing minor relative movement, so the two do not observe geometry from an identical pose. The offset is small against the room scale, and which architecture sat on top was not held constant, so it affects both symmetrically. The pair was also not carried by one person throughout (the researcher for roughly fifteen minutes, a second person for about ten, then the researcher again), so pace and gait vary within a run. Both headsets move together, so any change applies to both at once, but the motion path is not reproducible between runs, so comparisons are always made within one run.

Telemetry was captured by a purpose-built logging component writing a multi-row-type CSV every 100 ms: frame time and rate, battery percentage, voltage, current, battery temperature, OS thermal status, CPU and GPU performance levels, round-trip time, bytes transferred, and mesh vertex and triangle counts. Instantaneous power is voltage times current, which is independent of charge state and therefore preferable to battery percentage. The server logs per-stage GPU timings separately, joined to the client log by an echoed timestamp.

Runs began with both headsets at matched battery level and room temperature. **To separate architecture effects from hardware variance, the assignment of architecture to physical headset was swapped between runs.** Three runs were recorded across the two rooms in Table 3.1: the cluttered room twice, for 20 and 41 minutes with the architectures swapped for the second, and the hallway once for 18 minutes. Complexity is characterized



Figure 3.1: The dual-headset rig. Both Quest 3 units were mounted one above the other and carried as a single object, so that the two architectures observed the same room, motion and lighting at the same moment. The mounting is not rigid, which leaves the small vertical offset discussed above; which unit sat on top was not held constant between runs.

by mean vertex count rather than floor area, since vertex count drives reconstruction cost.

Room	**Approx. dimensions**	**Distance to access point**	**Mean vertices**
`cluttered_small`	3×5.8 m	1–3 m (same room)	37 000–42 000
`hallway`	2.3×25 m	4–10 m	55 000

Table 3.1: The two rooms. Dimensions are approximate. The access point stayed in one place; what changed was how far the headset traveled from it. The two rooms differ in both complexity and wireless distance at once, which limits what can be attributed to either alone.

3.3.2 User Study

Nine participants took part in two waves: P01–P05 between 15 July and 20 August 2026, and P06–P09 on 5 September 2026. Sessions ran 16 to 41 minutes. Participants were recruited by convenience from the researcher's social circle and received a protein bar as an incentive for participation. Sessions took place in four different rooms, listed in Table 3.2, with each participant using a single room throughout their session. In each room the

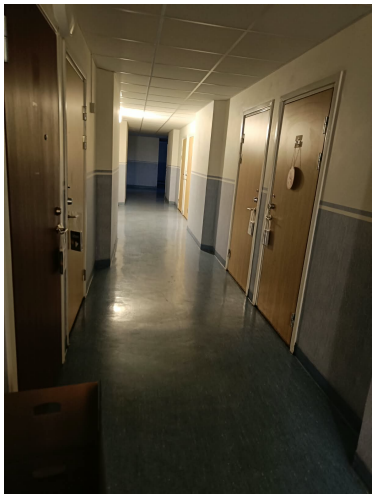


Figure 3.2: The `hallway` run. The space is long and narrow, holds about 30% more geometry than the cluttered room, and takes the headset 4–10 m from the access point. Both differences are present at once, which is why the hallway tests complexity and wireless distance together rather than separately.

participant stood in roughly 1×1 m of clear floor space. The wireless access point was between 1 and 3 m away in all four rooms.

Room	Participants	Distance to access point
1	P01, P02	1–3 m
2	P03	1–3 m
3	P04, P05	1–3 m
4	P06, P07, P08, P09	1–3 m

Table 3.2: Rooms used in the user study. Room geometry varied; distance to the wireless access point did not. Room 4 covers the whole of the second wave, so wave and room cannot be separated for those four participants.

The sample was four women and five men, seven aged 25–34 and two 35–44. Seven had used a headset "a few times", one "occasionally", one "regularly". **These figures were collected at two different times.** For P06–P09 the questionnaire collected them during the session. For P01–P05 no such instrument existed at the time, and the same details were collected from those participants directly, a few weeks after their sessions. Both sets are self-reported. The first wave's are retrospective. No participant characteristic is treated as a variable anywhere in this thesis.

After consent and a baseline symptom questionnaire, each participant completed four task rounds, an architecture comparison, three network rounds, and a network comparison.

In both tasks, participants held a virtual blaster and fired bullets. The blaster, the bullets and the targets were virtual and the room was real. In the **targets** task, floating virtual targets followed a fixed, repeatable path around the participant, who turned on the spot to track and fire at them.

In the **cover** task, the virtual targets were present and moving as before, and the researcher additionally moved a real opaque sofa cushion, approximately 40×50 cm, through the participant's field of fire every three to four seconds. **Participants were asked to treat the cushion as the priority and fire at it rather than at the targets.** Only the participant wears the headset, so the researcher could not see the virtual targets and the cushion was not aimed at any particular one; where a target was occluded by the cushion, that was incidental. The cushion is a real object, so it exists for the game only through the reconstruction: a shot registers against it only if the mesh has followed it. Because the mesh is hidden from the participant, whether shots stop where the cushion is, is the only evidence available. The manipulation was not described to participants. The cushion was moved by hand to the same instruction in every round, but the motion was not scripted and was not reproduced identically between conditions. **This cover task is the study's main discriminator.**

Counterbalancing

The two waves ran mirrored orders (Table 3.3), and this is central to reading the results. In wave 1, the order was fixed. Everyone began with on-device targets and ended with on-device cover, so the edge cover round always came first, and every architecture comparison was confounded with order, practice, and heat. Wave 2 inverted the sequence exactly. **If an effect shows up in both waves, the order did not cause it. If it shows up in only one, the order probably did.**

Because the rooms differed, room geometry is an uncontrolled variable in the user study, and no room identifier was recorded in the response files. The mapping in Table 3.2 comes from the researcher's session notes rather than from the data. Two things limit the damage. The distance to the access point, which the instrumented study shows is what actually determines whether the offloaded architecture keeps up, was the same in every room. And every comparison between architectures is made within a single

	Wave 1 (P01–P05)	Wave 2 (P06–P09)
Round 1	Targets — on-device	Targets — edge
Round 2	Cover — edge	Cover — on-device
Round 3	Targets — edge	Targets — on-device
Round 4	Cover — on-device	Cover — edge

Table 3.3: Task round order by wave; wave 2 is the exact mirror of wave 1. Two first-wave participants (P03, P05) took the target rounds consecutively, but the edge cover round still preceded the on-device one, so the direction of the confound was identical across wave 1. Counterbalancing between waves is weaker than randomizing per participant, but is enough to test whether an effect reverses when the order does.

participant's session, in one room, so a difference between rooms cannot produce one.

Network rounds

Three edge-only rounds per participant were run under induced degradation applied server-side with the macOS Network Link Conditioner, symmetrically to uplink and downlink (Table 3.4).

Condition	Bandwidth	Packet loss	Added delay	Used in
None	unthrottled	—	—	Wave 2
Moderate	20 Mbit/s	0%	25 ms	Both
Severe	10 Mbit/s	1%	50 ms	Wave 1
Extreme	5 Mbit/s	2%	100 ms	Both

Table 3.4: Induced network conditions. Each participant completed three rounds.

The condition set changed between waves, for a reason best stated before the results. The unconstrained architecture draws about 27–29 Mbit/s on the downlink (Chapter 5), so **every wave-1 condition capped bandwidth below what the system actually uses** — the mildest by roughly a third, the harshest by a factor of six. That block spanned three degrees of insufficiency rather than running from adequate to inadequate. Wave 2 replaced Severe with an unthrottled control, so the two waves are not poolable here and are reported separately in Section 5.4.6.

The participants saw only "Test 1" to "Test 3", so no severity-suggesting name could anchor their judgment. The mapping is in Appendix C. The intended per-participant randomization of that order did not occur in either

wave, four of five wave-1 participants received one sequence, and all four wave-2 participants another. Condition and position are therefore confounded within each wave, but the two sequences run in opposite directions, which allows an ordering effect seen in one to be checked against the other.

3.3.3 Instruments

All instruments were administered through a purpose-built web application on the researcher's device, storing one record per form.

Instrument	Content	Administered
Demographics	Age band, gender, prior head-mounted display use	Baseline (added for wave 2)
VRSQ	Nine symptoms, 0 (none) to 3 (severe)	Baseline and after each task round
NASA-TLX	Six subscales, 1–7, raw and unweighted	After each task round
S1	Reconstructed geometry looked accurate	Targets rounds
S3	System felt responsive to movement	All task and network rounds
S4	Reconstruction stayed stable while moving	Targets rounds
S5	Confidence shooting at targets behind cover	Cover rounds
R1	Game responded promptly when firing	Network rounds
D1	Round felt noticeably different from earlier ones	Network rounds
C1–C5	Forced-choice comparison across conditions	After each comparison block
Free text	Open responses on perceived differences	Network rounds and comparisons

Table 3.5: Questionnaire instruments.

One detail of administration should be recorded. Item S5 was worded "I felt confident shooting at targets behind cover", a residue of an earlier

version of the task. In every session the researcher clarified verbally, before the round, that the cushion was the priority and that the item referred to firing at it. Participants therefore answered about the interaction they had performed. The printed wording is reproduced unchanged in Appendix B because that is what was on screen, and the clarification is recorded here because it was given consistently to all nine participants.

Each instrument answers a different part of RQ4. The custom Likert items cover perceived reconstruction quality: S1, S3 and S4 ask whether the mesh looked accurate, felt responsive and stayed stable in the targets task, where nothing in the room moves; S5 asks whether the participant felt confident firing at the moving cover, which is the only item that depends on the reconstruction being correct about geometry that has just changed. R1 and D1 cover the network rounds, and test whether a degraded link is noticeable at all. NASA-TLX measures whether either architecture makes the task harder to perform, and VRSQ whether either makes participants unwell, since both are plausible consequences of a reconstruction that lags. The forced-choice items C1–C5 capture an overall preference, which is what a study reporting a single summary judgment would have collected on its own. The free-text prompts let participants describe a difference in their own words rather than only on a scale.

The resulting data is therefore **mainly quantitative**: ordinal self-report on fixed scales, reported per participant and by median. The free-text responses are the only qualitative component, and they are used to interpret the ratings rather than as evidence on their own.

Item S3 is deliberately repeated across all three contexts so the same construct can be compared between tasks.

3.4 Data Analysis

Telemetry was processed with Python. Power was derived from voltage and current and reported alongside battery discharge. Thermal behavior was analyzed as a trajectory over session time, to see whether an equilibrium was reached. Frame time is reported as a distribution (median, 95th and 99th percentiles) rather than a mean, because the difference between the architectures is in the tail, not the center. Network behavior was reported as sustained throughput each way and as round-trip time split into server computation and network transport.

User study data were reported **descriptively**. With nine participants and a design counterbalanced between waves rather than randomized, no

significance tests were reported. Per-participant values are given in full alongside medians, so the reader sees the individual data rather than a summary that hides the sample size. NASA-TLX is reported per subscale as well as a total, since the subscales diverge in ways the total hides. The study is powered to detect nothing. Its purpose is to establish the direction of user response.

Fourteen free-text responses were read in full and grouped by hand into themes. The volume did not warrant qualitative analysis software. Four were in Spanish and were later translated, with both versions in the appendix.

3.5 Ethical Considerations

The research follows standard ethical principles for computing research [17]. All participants received a written information sheet and signed a consent form before taking part.

Because the architecture processes depth data from the physical environment, specific privacy measures apply. **Depth data is processed entirely in volatile memory and never written to disk on either the headset or the server**; no image, video, or 3D model of the participant's surroundings is retained. In the edge condition, depth measurements cross a private local network to a machine in the same room and never touch the internet. These properties were disclosed in the information sheet.

Participant safety was a primary concern. Because motion sickness is a known risk, symptoms were measured before and after every round. Participants were told they could stop and remove the headset at any point without penalty, and the researcher stayed present throughout to prevent collisions. Data are identified only by a participant code, stored separately from the consent forms, kept for one year, and reported only in aggregate. Participants were given two weeks to withdraw their data after their session was conducted. No audio or video recording was made at any point. The free-text responses in Appendix D were typed by participants themselves.

The study did not require review by a research ethics board. It involved no sensitive personal data, no vulnerable groups, and no deception, so it falls outside the categories requiring formal approval. The procedures above were applied on standard principles rather than in response to a board's conditions.

The use of generative AI tools in this project is disclosed in Section 6.9.

4

Requirements

Both artifacts solve one problem: keeping a triangle mesh of a real room current enough that an application can collide with real geometry and be hidden behind it. This chapter sets out what any solution must do, wherever the work is done. These requirements are the benchmark used in Chapter 5.

4.1 Where the requirements came from

Following the Design Science principle that requirements come from existing knowledge and from the problem rather than from the designer [8], they were drawn from three sources.

From existing research. KinectFusion and its successors define TSDF fusion and the closed-surface property physics engines need [9, 16], and Chisel shows it is practical on mobile hardware [13]. The mobile XR literature sets the limits: motion-to-photon delay in the low tens of milliseconds, within a few watts [15, 10]. The edge computing literature adds that offloading only works when the network costs less than the computation it replaces [14, 3, 20].

From the problem. Consumer devices scan once and lock the result, so the mesh never changes. A bullet behaving correctly around a door that has just opened needs the mesh to follow the room. FR1 and FR5 come from this.

From building the artifacts. Two requirements emerged from piloting rather than reading, and each came from one of the research strategies in Chapter 3. Pilot instrumented runs (the *experiment* strategy) showed the artifacts could not be compared unless their telemetry shared a format and a clock, which became NFR10. Pilot questionnaires (the *survey* strategy) showed that participants who could see the mesh judged its appearance rather than the interaction, which is why it is hidden. So the requirements taken from the literature preceded the artifacts, while these two came out of the design cycle itself.

4.2 Functional requirements

ID	Requirement
FR1	The system shall rebuild the room as a triangle mesh *continuously*, updating as the camera moves and as objects move, rather than once during setup.
FR2	The mesh shall form a closed, manifold surface a physics engine can use as a collider.
FR3	Real geometry shall hide virtual objects behind it: a target placed behind a real object must not be visible through it.
FR4	Bullets shall stop against real geometry, including against an object that is moved during the session.
FR5	Objects moved into or out of the room during a session shall appear in the mesh without a rescan.
FR6	Both architectures shall use identical game logic, scene content and reconstruction math, differing only in where the work is done.
FR7	The system shall record per-frame telemetry detailed enough to reconstruct performance afterward: frame time, temperature, battery, network volume, round-trip time and mesh size.

Table 4.1: Functional requirements.

FR6 carries the most weight, because it is what makes the comparison trustworthy. Both artifacts use the same TSDF formulation, the same frustum-point optimization, and the same Naive Surface Nets meshing; the Metal code on the server is a direct port of the HLSL shader on the headset. The only thing changed is where the heavy work happens. One difference

follows from that rather than adding to it: the offloaded path updates at about 10 Hz instead of 15 Hz, because it cannot send faster than one round trip allows (Section 3.2.1).

4.3 Non-functional requirements

ID	Property	Requirement
NFR1	Steady frame rate	Rebuilding the room must not drag rendering below the display refresh rate, nor cause single frames slow enough to break the motion-to-photon budget.
NFR2	Heat	The headset must not start throttling within a session of realistic length, taken here as 20 minutes.
NFR3	Power	Battery use must allow a session of 20 minutes.
NFR4	Freshness	The delay between something moving and it appearing in the collider mesh must stay below the point where the interaction feels wrong.
NFR5	Bandwidth	The offloaded architecture's network use must fit inside ordinary local Wi-Fi.
NFR6	Graceful degradation	When the network worsens, the offloaded architecture must degrade gradually rather than stop working.
NFR7	Spatial accuracy	The mesh must match the real room closely enough that users notice no mismatch.
NFR8	Independence	The on-device architecture must work with nothing external; the offloaded one's dependency must be stated and bounded.
NFR9	Privacy	Depth and camera data must stay in volatile memory and never be written to disk, in either architecture.

NFR10	Comparability	Both artifacts must write telemetry in the same format, joinable across the client–server boundary by timestamp.

Table 4.2: Non-functional requirements. The groups map onto the research questions: NFR1–NFR3 to RQ1, NFR4–NFR5 to RQ2, NFR6 to RQ3, NFR7 to RQ4, and NFR8–NFR10 are conditions the comparison itself depends on.

One limit is stated here rather than discovered later. **NFR4 has no established threshold.** Nothing in this thesis determines the delay at which an interaction starts to feel wrong, and the literature does not supply one for this task. The delay is therefore measured and reported rather than judged pass or fail.

NFR9 is the one requirement with an ethical consequence. Depth and camera frames stay in volatile memory in both architectures. In the offloaded case, they cross the network but are stored at neither end. These are checkable by inspection and by what participants were informed.

4.4 Mapping to the research questions

RQ	**Requirements tested**	**Evidence**
RQ1	NFR1, NFR2, NFR3	Instrumented study
RQ2	NFR4, NFR5	Instrumented study
RQ3	NFR1, NFR3, NFR6	Instrumented study
RQ4	NFR4, NFR7	User study
—	FR1–FR7, NFR8–NFR10	Artifact inspection

Table 4.3: Requirements mapped to the research questions.

The last row is different from the others. Those requirements are not hypotheses being tested, they are conditions that had to hold before the comparison was worth running, and they are confirmed by inspecting the artifacts rather than by measurement.

5

Evaluation and Comparison

This section identifies the measurements for each artifact in line with the efficiency of each technology, as detailed in Chapter 4. It gives the findings without interpretation; what they mean is discussed in Chapter 6.

5.1 Evaluation Strategy

In Design Science terms, this kind of evaluation can best be described as an *ex post, artificial* evaluation. This means that, in this case, both artifacts were finished before evaluation, rather than being evaluated in use, in operations, or in everyday settings.

The evaluation also has two parts, corresponding to the two research studies in Chapter 3.

The **instrumented study** is the first measured part. In this study, two headsets were placed vertically and physically combined, carrying the two architectures for the same room at the same time. Data recorders attached to the software application captured power consumption, headset battery temperatures, frame timing, and wireless network load levels. This section addresses RQ1, RQ2, and the measurable part of RQ3.

The **user study** covers the user experience. Nine subjects rated the two architectures while performing two tasks under three different network conditions. Two groups were used. The second group ran the experiments in the opposite order compared to the first. This study answers RQ4.

5.2 Functional Verification

Both artifacts meet all seven functional requirements. Both rebuild the room continuously (FR1), produce surfaces the physics engine can use (FR2), hide virtual content behind real objects (FR3), stop bullets against real geometry (FR4), and pick up objects that are moved during a session (FR5). Both come from one Unity project with the same game logic and the same reconstruction math (FR6), and both write the same telemetry format (FR7).

There are two points worth discussing.

Initially, FR3 failed right away in the edge configuration. The server returned mesh chunks that were not incorporated into the collision layer, and the virtual targets were still visible through the physical objects. This issue was already addressed before the final evaluation builds started.

Second, both artifacts comply with FR5 but not to an equal degree. When an object is moved into or out of the room, the edge path takes a visible moment to catch up. The on-device path does not. That delay is exactly what the cover task in Chapter 3 was built to reveal.

5.3 Instrumented Study Results

Three experiments were done and captured for the last section, which are summarized in Table 5.1. The first two took place in a cluttered room: a 20-minute study and a 41-minute study during which the headsets were exchanged. The exchange mattered, and Section 5.3.2 explains why. The third experiment was in a hallway, where the space is more complicated and also meets the condition of being significantly farther from the Wi-Fi access. Every study began with both headsets' battery life set to the same levels and at room-level temperature.

Run	Room	Duration	Mean vertices (on-device)
1	`cluttered_small`	20 min	37 123
2	`cluttered_small`	41 min	42 208
3	`hallway`	18 min	55 210

Table 5.1: The three instrumented runs. Geometric complexity is measured by the mean vertex count of the on-device reconstruction, because vertex count is what drives the cost of rebuilding the room.

By this comparison, the hallway holds about 30% more geometry than the cluttered room. Despite this fact, it is also surrounded by the worst

wireless conditions, as Section 5.3.4 shows. Therefore, it evaluates both aspects of RQ3 together.

5.3.1 Heat (NFR2)

Neither architecture ever throttled under normal operating conditions. The operating system thermal status remained at 0 for every minute of all six recordings. Table 5.2 shows battery temperature over the 41-minute run.

Elapsed (min)	On-device (°C)	Edge (°C)
0	41.6	40.9
10	49.0	49.5
20	49.1	50.3
30	49.3	51.6
40	50.0	52.6

Table 5.2: Battery temperature over the 41-minute run in `cluttered_small`.

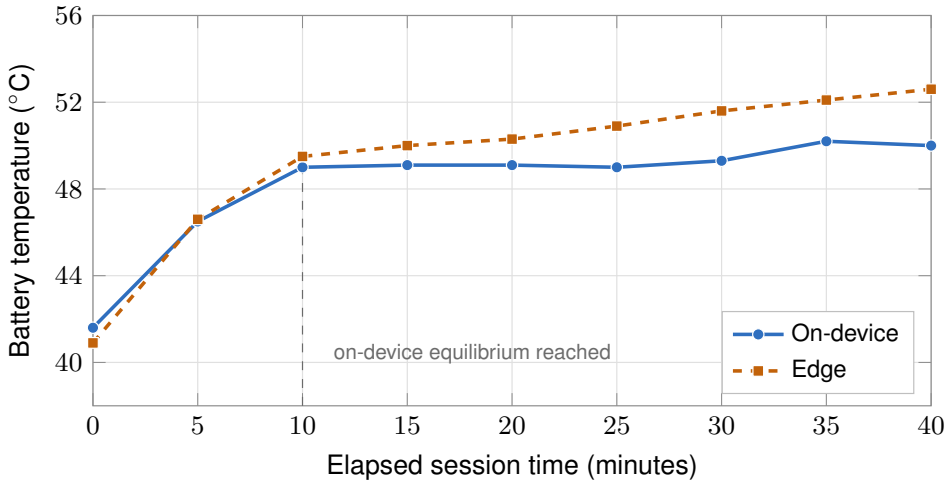


Figure 5.1: Battery temperature over the 41-minute run in `cluttered_small`. The on-device architecture reaches thermal equilibrium at approximately 49 °C after ten minutes and holds it for the remainder of the session; the edge architecture runs two to three degrees hotter and is still climbing at the end of the run. Neither architecture triggered the operating system's thermal governor at any point.

The on-device architecture tends to normalize to about $49\,^{\circ}$C after ten minutes and holds at that for the next thirty (Figure 5.1). The system is producing heat and losing heat at the same rate, well below the temperature at which the operating system would step in and throttle. The edge structure settles two to three degrees higher and keeps on moving up until the end of the session.

Frame rate does drop a little over the session, from a mean of 58.5 to 54.6 fps on-device. However, there was never a report of operating-system throttling, so that drop must have been due to another influence.

The hallway run reached $47\,^{\circ}$C on-device and $49\,^{\circ}$C on edge across 18 minutes. That follows the same pattern, even though the hallway contains more geometry.

5.3.2 Power (NFR3)

Table 5.3 presents mean power, derived from sampled voltage and current, alongside the battery consumed in each run.

The two headsets are different physical devices, each with its own battery and its own thermal behavior. To stop that from contaminating the comparison, the architectures were deliberately swapped between the headsets after the first run. Table 5.3 records which unit ran which architecture, and the results cannot be read without it.

Run	Room	Unit / architecture	Mean power (W)	Battery (pp)
1 (20 min)	cluttered	Unit 1 — On-device	9.66	17
1 (20 min)	cluttered	Unit 2 — Edge	10.95	21
2 (41 min)	cluttered	Unit 1 — Edge	10.04	38
2 (41 min)	cluttered	Unit 2 — On-device	10.34	41
3 (18 min)	hallway	Unit 1 — Edge	10.36	17
3 (18 min)	hallway	Unit 2 — On-device	10.10	17

Table 5.3: Mean power and battery use, showing which physical unit ran which architecture. The architectures were swapped between units after the first run.

Simply comparing the two rows of a single run would be misleading (Figure 5.2). Each run compares two architectures, but it also compares two different headsets, and one run alone cannot distinguish between them.

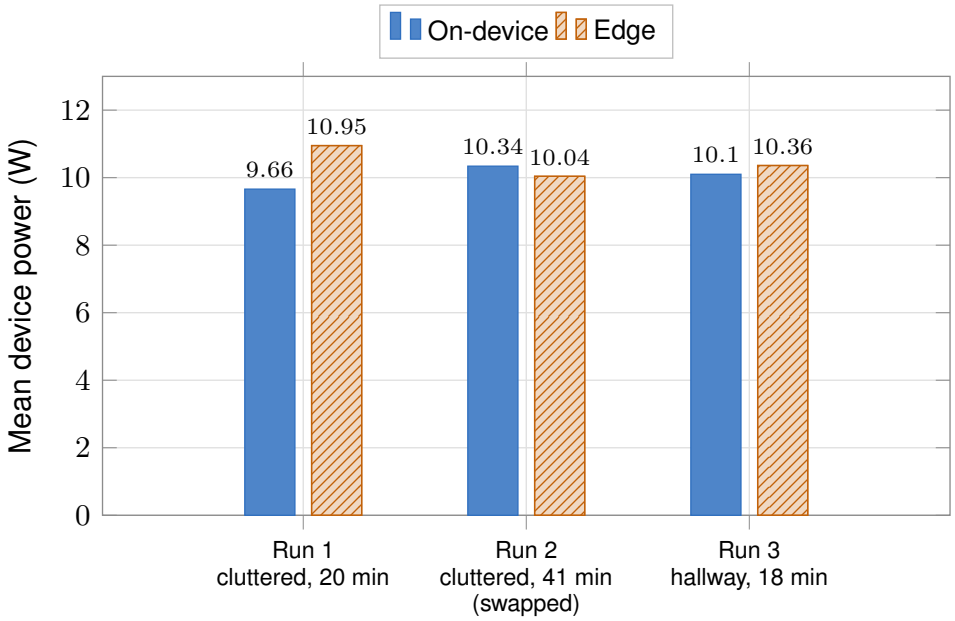


Figure 5.2: Mean device power recorded in each run. These raw figures are *not* a valid architecture comparison: each run pairs two different physical headsets as well as two architectures, and the hardware difference between the units (0.79 W) is larger than the architecture difference it hides. The architectures were swapped between units after the first run, which is why the apparent direction changes. See Table 5.4 for the crossover analysis that separates the two effects.

Runs 1 and 2 solve this. They were recorded in the same room with the architectures swapped, so between them each headset ran each architecture exactly once. That is a 2×2 crossover, and it lets the two effects be separated.

	Unit 1	**Unit 2**	**Mean**
On-device	9.66	10.34	10.00
Edge	10.04	10.95	10.49
Mean	9.85	10.64	

Table 5.4: Crossover analysis of mean power (W) in `cluttered_small`. Each unit ran each architecture once, so the row means give the architecture effect and the column means give the hardware effect.

There are two effects, and they are about the same size.

- The **hardware effect** is 0.79 W. Unit 2 always draws more power than Unit 1, whatever it is running.
- The **architecture effect** is 0.49 W. On the same headset, edge draws more than on-device.

Since the hardware impact is bigger, any comparison that leaves it out might not be reliable. In the raw run results, you can see this right away. In run 1, edge looks about 13% worse. In run 2, after swapping the headsets, edge looks better.

If you compare using just one headset, the hardware mismatch issue goes away. Then both units give the same result. This holds for all three runs:

- **Unit 1:** 9.66 W on-device, against 10.20 W on average when running edge — 0.54 W more, or 5.6%.
- **Unit 2:** 10.22 W on average on-device, against 10.95 W when running edge — 0.73 W more, or 7.1%.

Offloading therefore costs about 0.5 W more, close to 5%, than doing the work on the headset. That is the opposite of what the architecture was built to achieve.

This figure is more careful than being generous. In runs 1 and 2, a server bug caused many mesh updates from the edge setup to be discarded (see Section 5.3.5). The headset never paid for the dropped work. If the edge path had been working end-to-end, it would have used more power, not less.

5.3.3 Frame Time (NFR1)

Run	Architecture	p95 (ms)	p99 (ms)	>20 ms	>33 ms
20 min	On-device	35.2	47.4	27.9%	11.0%
20 min	Edge	27.9	181.9	12.6%	1.5%
41 min	On-device	34.9	46.3	—	—
41 min	Edge	27.8	223.7	—	—

Table 5.5: Frame-time distribution. Edge is steadier most of the time but has a far worse worst case.

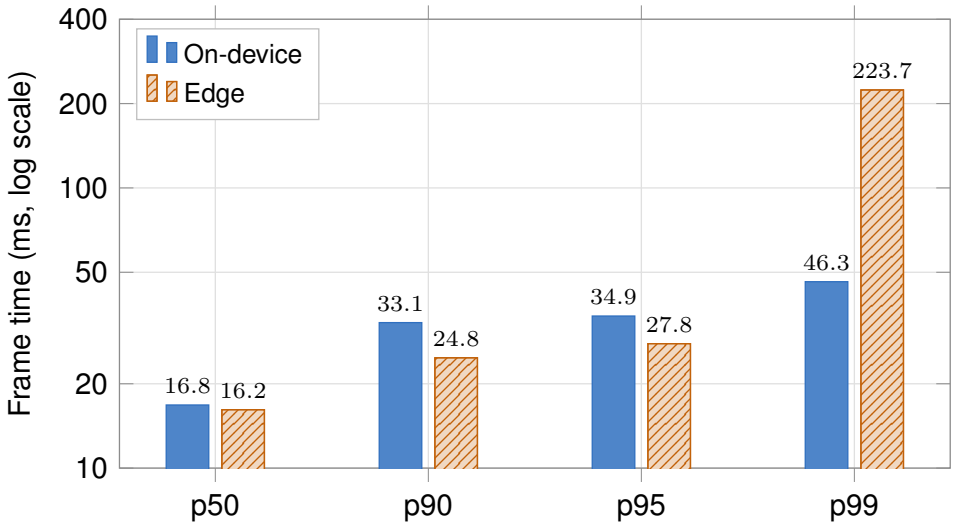


Figure 5.3: Frame-time distribution over the 41-minute run. The edge architecture is more consistent through the bulk of the distribution — lower at the 90th and 95th percentiles — but its 99th percentile reaches 224 ms against the on-device architecture's 46 ms. The two architectures fail in different ways: one produces many moderately long frames, the other produces fewer but far more severe stalls. Note the logarithmic axis.

The two architectures go wrong in different ways (Figure 5.3).

On-device produces many moderately slow frames. More than a quarter of all frames take longer than 20 ms.

Edge produces far fewer slow frames, but its worst frames are much worse. At the 99th percentile it reaches 182 ms in one run and 224 ms in the other. Those are single stalls long enough for a user to feel as a jolt.

One number in the hallway run cuts against the rest. The offloaded architecture averaged 70.4 fps there and missed the display refresh on only

47% of frames, its best frame-time result of the three runs. It achieved that because its mesh pipeline had fallen to 0.92 updates per second and had almost nothing left to do. Frame rate on its own is therefore a poor indicator of whether a reconstruction system is working.

5.3.4 Bandwidth and Latency (NFR4, NFR5)

	cluttered_small		hallway
Measure	**20 min**	**41 min**	**18 min**
Uplink (depth sent)	6.57	6.06	**1.52**
Downlink (mesh returned)	29.29	26.71	**3.71**
Round-trip time, median (ms)	105	94	**651**
Round-trip time, 95th pct. (ms)	297	316	**2 721**
Server computation, median (ms)	4.78	4.66	5.03
Mesh updates delivered	5 021	9 113	**1 021**

Table 5.6: Network demand and where the latency goes, for the edge architecture. Throughput is in Mbit/s. The hallway column is emphasized because it is a change in kind, not a change in degree.

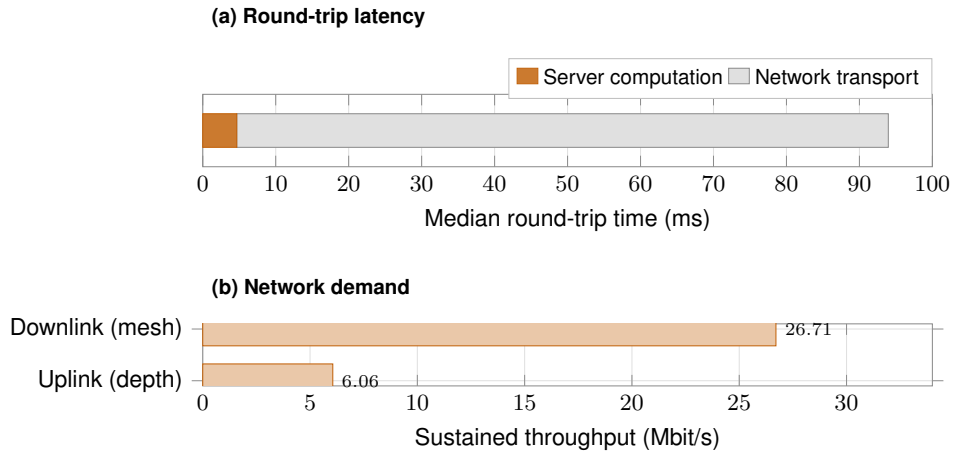


Figure 5.4: Where the cost of offloading actually falls, measured over the 41-minute run. **(a)** Of a median 94 ms round trip, the server spends under 5 ms computing; the remaining 95% is network transport, so a faster server would change almost nothing. **(b)** The reconstructed mesh returning to the headset is roughly 4.5 times larger than the depth stream that produced it, making the downlink rather than the uplink the binding constraint.

Three results stand out (Figure 5.4).

First, the return trip carries more data than the outward one. In the cluttered room the mesh stream that comes back is about 4.5 times the size of the depth stream that made it. Over 41 minutes, this totaled 8.2 GB.

Second, the delay is almost all network, not computation. In all three runs, the server completes its own work in around 5 ms. In the cluttered room, that is about 5% of a 94 to 105 ms round trip. In the hallway, it is below 1% of a 651 ms round trip. **The server is not the limiting factor**. A quicker server would not change the outcome.

Third, and most importantly, the edge architecture failed in the hallway. The median round trip jumped by six times to 651 ms. The 95th percentile went past 2.7 seconds. Long run throughput fell to about a quarter of what the cluttered room reached.

The last row of Table 5.6 shows what this looked like. The edge setup delivered 1 021 mesh updates in 18 minutes, which is just under one per second. In the cluttered room, it reached about four per second. Since the mesh is only as current as its last update, the collision geometry in the hallway was badly out of date.

This breakdown comes from the one-request-at-a-time rule in Chapter 3. The client waits for each reply before it sends again, so it can never update faster than one round trip allows. With a 105 ms round trip, that ceiling is about nine updates per second, which is high enough not to matter. With 651 ms it drops to about 1.5 per second, and the measured 0.92 fits that. The round trip itself grew because the link got worse, most likely because the hallway was further from the router and the signal was weaker. **So the limit in the hallway was the waiting for each reply, not the amount of data.**

The on-device architecture did not change here. It never uses the network at all. This is NFR8, independence from infrastructure, shown as a measurement instead of a design claim.

5.3.5 Robustness (NFR6)

A bug in the server mesh maker created some chunks that referenced vertex indices that didn't exist. In other words, the indices pointed past the end of the vertex array for that chunk. The client noticed and dropped those chunks instead of trying to draw them. Over 20 minutes, this led to 1 014 rejected chunks, or 16.8% of replies. Over 41 minutes, the count rose to 4 562, or 33.4%.

The root issue was a buffer that was too small. The meshing step set the buffer assuming it would only need one vertex for every three cells in a chunk. That plan also assumed that no more than one third of cells would ever touch the surface. Some cluttered parts of the environment did not fit that assumption. Then the kernel wrote more data than the buffer could hold. After that, the host reported a vertex count smaller than the indices already written.

Later, it was fixed by sizing the buffer for the worst case instead. Surface Nets produces at most one vertex per cell, so that is the worst case. Bounds checks were added to both kernel passes.

The fix worked: After the change, the hallway test showed zero rejections across 1 021 replies.

The two earlier runs are still not clean. That is why the direction of the bias matters. The study recorded a 41-minute run and split it into one-minute blocks. Then grouped the blocks by the number of rejections in each block. Those results are shown in Table 5.7.

Window type	Frames	p50 (ms)	p95 (ms)	p99 (ms)
Zero rejections	11 689	13.98	27.83	28.94
Above-median rejections	58 018	16.61	27.84	224.14

Table 5.7: Edge frame times in the 41-minute run, split by how many rejections occurred in each one-minute window.

The worst frame times occur in almost the same blocks with the most rejections. When rejections happen often, the 99th percentile frame time is 224 ms. With no rejections, the 99th percentile drops to 29 ms.

Both numbers come from the same story. A chunk got rejected only when its region contained more surface than the buffer could store. So the rejections line up with the room's densest areas. Those dense areas also create the biggest meshes. They also trigger the longest collider rebuilds.

This suggests the bias shifts the data. If the rejected chunks had been accepted, they would have added mesh and physics work exactly in the blocks that already look like the worst stalls. **The edge frame-time tail reported here is therefore too low, not too high.** The same reasoning applies to power. Work that was discarded was work the headset never paid for, so a fully working edge path would draw more power than reported here, not less.

So both main findings, that edge uses more power and stalls worse, are cautious ones. A working edge path would make them stronger, not weaker. That is why these two runs are still usable despite the defect.

5.4 User Study Results

This section reports what participants rated, and answers RQ4. Every value is given per participant as well as by median, because with nine people a median alone would hide the sample size.

5.4.1 Participants

Nine people, labeled P01 to P09, finished ten questionnaires each. This included one at the start, four after task rounds, three after network rounds, and two for comparisons. Out of all ninety forms, every one had complete answers. Each session lasted from 16 to 41 minutes.

The sample included four women and five men. Seven people were between 25 and 34 years old. Two people were between 35 and 44 years old. Seven used a headset “a few times”. One used it “regularly”. Sessions were run in four different rooms rather than one (Table 3.2), with each participant in a single room throughout. All four rooms placed the wireless access point within 1–3 m of the participant. One used it “occasionally”. As explained in Chapter 3, the second wave answered these items during the session. The first wave was asked the same questions a few weeks afterward, so their answers are self-reported but retrospective.

The two waves handled the task rounds in opposite order. Table 3.3 shows this. For P01 to P05, the edge cover round came before the on-device round. For P06 to P09, the order was flipped. The results below are split by wave when order could matter. This is because agreement in both waves is the only sign that an effect is not just due to where it appeared.

5.4.2 Perceived Quality by Task

Table 5.8 lists the custom Likert scores. Each participant has one value. The values are shown in the order P01 through P09. The scale runs from 1 to 7. Here, 1 means strongly disagree. A 7 means strongly agree.

The targets task cannot tell the architectures apart. In both conditions, all three items are at about the top end.

Task	Item	On-device	Md	Edge	Md
Targets	S1 geometry accurate	6, 7, 6, 7, 7, 7, 7, 5, 7	7	7, 7, 6, 7, 7, 6, 7, 7, 7	7
Targets	S3 felt responsive	6, 5, 7, 7, 7, 7, 5, 5, 7	7	7, 6, 7, 5, 7, 7, 7, 6, 7	7
Targets	S4 stayed stable	7, 6, 7, 7, 7, 7, 5, 5, 7	7	7, 6, 7, 7, 7, 7, 6, 5, 7	7
Cover	S3 felt responsive	4, 5, 2, 7, 7, 7, 4, 7, 4	5	6, 5, 5, 7, 7, 7, 6, 6, 6	6
Cover	S5 confident shooting	5, 6, 3, 5, 5, 4, 4, 7, 5	5	**7, 7, 5, 7, 7, 7, 5, 7, 7**	**7**

Table 5.8: Custom Likert answers by task and architecture, participants P01–P09 in order.

The cover task can. Item S5 asks how confident the participant felt while firing at the moving cover. Eight of nine people score higher under edge. The ninth person is tied. **No one is lower in edge (Figure 5.5).**

S5 is not caused by the order of the rounds

S5 is the one clear architectural pattern in the whole study. For the first wave, it was hard to trust the numbers. In that wave, the edge cover round always happened first. So practice, fatigue, and headset temperature may have helped the edge. In the second wave, the order was intentionally changed. Table 5.9 shows the two waves in separate parts.

Wave	Cover order	On-device	Edge	Favors
1 (P01–P05)	edge first	5, 6, 3, 5, 5	7, 7, 5, 7, 7	edge $5/5$
2 (P06–P09)	on-device first	4, 4, 7, 5	7, 5, 7, 7	edge $3/4$, tie 1
All nine		**Md 5**	**Md 7**	**edge** $8/9$**, tie** 1**, on-device** 0

Table 5.9: Item S5 by wave. The second wave met the two cover rounds in the opposite order to the first.

The effect survived the reversal. In the wave that met the on-device cover round first, three of four rated edge higher and the fourth rated them equal. When the order changed, nobody picked the device option as better.

If fatigue or practice drove the pattern, flipping the order should have flipped what people chose. That did not happen. So the order itself is not the cause. That leaves the system design, which is discussed in Chapter 6.

In the cover task, Item S3 points in the same direction, but the signal is smaller. Edge was higher for four people, tied for four, and lower for one across both waves.

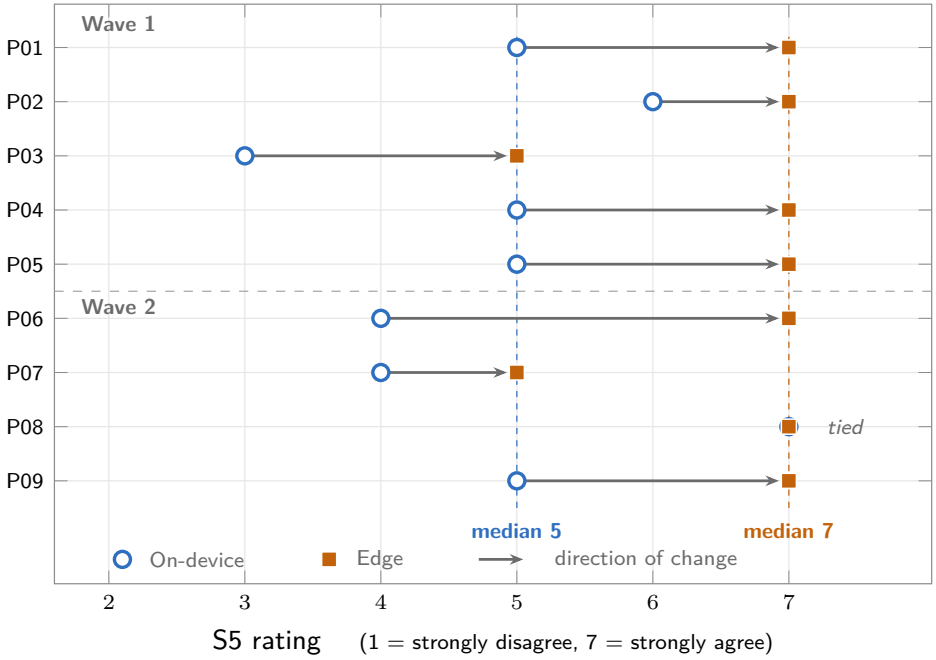


Figure 5.5: Per-participant ratings on item S5 (confidence firing at the moving cover), on-device against edge, for all nine participants. Each arrow runs from that participant's on-device rating to their edge rating, and every arrow points the same way. P08 rated the two equally and therefore has no arrow; no participant rated on-device higher. Wave 1 met the edge cover round first and wave 2 met the on-device cover round first, so the effect holding in both is what rules out presentation order (Table 5.9). Dashed lines mark the condition medians.

5.4.3 Workload

Table 5.10 lists the raw NASA-TLX totals. This does not include the Performance subscale in any total for the reasons below. Each sum uses five subscales, so scores run from 5 to 35.

Task	Architecture	Per participant (P01–P09)	Median
Targets	On-device	7, 7, 18, 8, 15, 7, 11, 10, 5	8
Targets	Edge	6, 5, 10, 9, 13, 6, 9, 14, 6	9
Cover	On-device	14, 5, 19, 7, 13, 8, 11, 8, 7	8
Cover	Edge	6, 8, 22, 8, 17, 8, 11, 7, 5	8

Table 5.10: NASA-TLX totals, five subscales, Performance excluded.

Workload stayed low in all cases. None of the conditions even gets to the middle of the scale. The designs also do not show clean separation. The medians differ by about one point or less in each check, while people's scores vary much more than the differences between conditions.

Why the Performance subscale was dropped

In the usual NASA-TLX setup, the Performance scale is reversed. On that scale, 1 is best, and 7 is a failure. In this study, all other parts go the normal way, so 7 means good.

In the first wave, the standard version was used. The responses suggest people read it wrong. For the target rounds, the median for Performance was 5, which is near failure. Yet the same people rated accuracy, speed, and stability at 7. They also gave 1 for frustration. Someone who truly thought they failed would not choose the very lowest frustration at the same time as the very highest satisfaction.

In the second wave, the item was changed. It was set to run from Failure to Perfect in the survey view. Then, when the answers were saved, it was flipped back to match the standard direction. With that update, the pattern fits. People judged their own shooting as only moderate, but they still rated the system as strong. Those two views can both be true, meaning the reconstruction likely worked and the participant did not land the shot. This is also why the first wave data cannot be used to spot the same effect.

Because the subscale was asked in two different forms and one was likely misread, those were left out from all totals. Appendix C still shows it for both waves. The first-wave values are flagged as unreliable.

5.4.4 Simulator Sickness

VRSQ scores (nine questions, total 0 to 27) remained at the bottom in both round sets.

Out of the nine, seven people completed all rounds with scores at or below 1. Three of those seven (P01, P06 and P09) began above zero and ended at or below their baseline.

Only two people mentioned anything that stood out. For P03, signs showed up during the cover rounds, and they showed up for both setups. The counts were 8 on device and 6 for edge. P07 began with a score of 4. After that, scores stayed between 3 and 5 in every round.

A careful read is the right choice here, and the wording should stay narrow. Still, the base now has nine people, not five. With these round lengths, most people did not feel unwell. When symptoms did show up, they did not lean toward either architecture. Round times were about 1.5 to 7 minutes. That is also the window where VRSQ is least likely to pick up signs.

One detail is worth noting, even if it is not a point to stretch. The two who reported symptoms are also the most experienced headset users in the group. P03 and P07 are the only ones who said they use the headset “regularly” and “occasionally”. Everyone else said “a few times”. P03 also gave the lowest confidence scores on the cover task for both architectures. P03 also had the highest workload totals in three of the four rounds.

Two people cannot settle whether experienced users judge a reconstruction more harshly, or simply notice their own symptoms sooner. This is written down because this sample has almost no experienced users. A follow-up that intentionally recruited experienced users might not match what is seen here.

5.4.5 Direct Architecture Comparison

Item	01	02	03	04	05	06	07	08	09	D/E/B
C1 most responsive	E	B	D	B	D	D	E	E	B	3 / 3 / 3
C2 30-minute session	E	D	D	B	D	E	E	E	B	3 / 4 / 2
C3 best reconstruction	B	B	E	D	D	D	E	E	B	3 / 3 / 3
C4 most comfortable	E	B	D	B	D	D	E	E	B	3 / 3 / 3
C5 would recommend	E	B	D	B	D	D	E	E	B	3 / 3 / 3

Table 5.11: Forced-choice architecture comparison, P01–P09. D = on-device, E = edge, B = both / no difference noticed.

Neither architecture wins any item. Out of five, four items split into three groups with no strong tilt. The remaining one, C2, runs as a 3 to 4 to 2 split. With nine people, this is about as close to no preference as the data can show.

Read on its own, the first wave alone, it looked different. Three out of five picked on-device for the long session. That looked like a real preference in the study. Then the second wave moved the other way on the same item, and most of that choice went away. This was treated as noise. The item never had a steady majority from the start.

Two answer patterns matter here. They limit what these items can say. P07 and P08 pick the edge option on all five items. P09 picks "both" on all five items. When a person answers the same way across five questions with different wording, it can feel like one repeating impression. It is not five separate judgments. Items C1, C4 and C5 make the point sharpest: every participant gave all three the same answer, so three differently worded questions produced one identical set of responses.

5.4.6 Network Degradation Rounds

Each participant also completed three short rounds using edge only. These rounds used different levels of forced network throttling. They were just named "Test 1" through "Test 3". Table 3.4 lists the exact setups. Bandwidth was capped in each direction at 20, 10, and 5 Mbit/s. Packet loss was set to 0, 1, and 2 percent. Delay was set to add 25, 50, and 100 ms.

Item	Moderate	Md	Severe	Md
R1 responded promptly	6, 7, 3, 4, 7	6	5, 5, 7, 2, 6	5
S3 felt responsive	6, 7, 3, 4, 7	6	6, 5, 5, 2, 6	5
D1 noticeably different	2, 7, 7, 7, 7	7	3, 6, 5, 7, 6	6
Extreme: R1 = 5, 3, 3, 7, 7 (Md 5)	S3 = 6, 4, 3, 7, 7 (Md 6)		D1 = 2, 7, 6, 7, 4 (Md 6)	

Table 5.12: Network round answers by actual condition, P01–P05.

Wave 1 — every condition was starved. The effect is small and in the expected direction. Moderate scores come out highest on both responsiveness items. Still, Severe and Extreme do not differ from each other on any item. The spread across participants is also wide.

This comes from the network settings, not from what people could sense. The unthrottled case uses about 27 to 29 Mbit/s on the downlink (Section 5.3.4). Moderate then caps at 20. Severe caps at 10. Extreme caps at 5. In this block, each condition starves the return path.

When the downlink is already low, lowering it again changes how far the mesh falls behind. It does not change whether it falls behind. The participant is looking at a stale reconstruction in all three cases. The block covers three grades of too-slow, rather than running from fast enough to too slow.

Wave 2 — with an unthrottled control. The second wave replaced Severe with an unthrottled round, giving the block something to compare against.

Item	None	Md	Moderate	Md	Extreme (Md)
R1 responded promptly	7, 6, 6, 5	6.0	7, 5, 7, 5	6.0	**4.0**
S3 felt responsive	7, 6, 6, 5	6.0	7, 5, 7, 5	6.0	**4.5**
D1 noticeably different	6, 7, 6, 5	6.0	6, 7, 4, 5	5.5	4.5

Table 5.13: Network round answers by actual condition, P06–P09. Extreme values per participant: R1 = 7, 4, 4, 3; S3 = 7, 4, 5, 4; D1 = 6, 6, 2, 3.

This time the conditions separate. Extreme sits about two points below the other two on both responsiveness items.

The bandwidth numbers match that. The Moderate limit at 20 Mbit/s sits near the usual setup. So people would not notice the change. Extreme at 5 Mbit/s is different, and it shows.

Position no longer decides the answer. In the first wave, converting each participant's final choices back to actual conditions revealed an unexpected pattern: **Four out of five chose their first network round as best**. This held even though the first round could match different setups. The one person who did not follow that rule was also the only one who chose the least degraded setup overall. It looked like position, not setup, was driving the decision.

In the second run, all four people saw the harshest setup first. That is the strictest test of the position explanation. Still, three of the four did not pick it. P07 and P09 chose the unthrottled round. P08 chose Moderate. P06 leaned a bit toward the starting round, but their scores were flat. P06 gave every round a 6 or 7 and did not separate them.

So the first run does not repeat when the first round is the worst. A simpler view fits both runs. People can name a link as worse only when the

drop is strong enough. The first run missed that cue because none of its three starting setups was fully unthrottled.

5.4.7 Summary of User Study Results

1. The targets task is at the top of the scale for both architectures in both waves and cannot tell them apart.
2. The cover task can. It favors edge on confidence for eight of nine participants, with the ninth equal. This is the opposite of what the edge path's latency would predict.
3. **That result is not caused by the order of the rounds.** It holds in both waves, including the one that met the on-device cover round first.
4. Workload is low and does not separate the architectures. The Performance subscale was misread in the first wave, fixed for the second, and left out of all totals.
5. Simulator sickness is at the floor for seven of nine participants, and for the other two it does not favor either architecture.
6. Overall preference does not follow the per-round ratings. Four of five forced-choice items split exactly three ways across nine participants. Neither architecture wins.
7. Network conditions separate once an unthrottled round is included. The harshest condition scores about two points below both the mild and the unthrottled ones, which are indistinguishable from each other.
8. The first wave's apparent first-round bias does not repeat in the second, where the harshest condition came first and three of four participants did not choose it.

5.5 Requirements Comparison

ID	Architecture A (On-device)	Architecture B (Edge)
FR1–FR2	Met	Met
FR3	Met	Met after the chunk-store fix
FR4	Met	Met

ID	Architecture A (On-device)	Architecture B (Edge)
FR5	Met, with no delay	Met, with a visible delay
FR6–FR7	Met by construction	Met by construction
NFR1	Not met: mean 54.1–56.8 fps against a 72 Hz display; p95 35 ms, p99 46–47 ms	Not met: mean 59.8–60.5 fps in the cluttered room; p95 28 ms, p99 182–224 ms
NFR2	No throttling; settles ≈ 49°C	No throttling; 2–3°C higher
NFR3	9.66–10.34 W	$+0.5$ W ($\approx 5\%$) on the same unit
NFR4	Not applicable	RTT median 94–105 ms; 95% is network
NFR5	Not applicable	6 Mbit/s up, 27–29 Mbit/s down
NFR6	Not applicable	Hallway: 4/s $\rightarrow$ 0.92/s. Degrades, but sharply
NFR7	S1 median 7	S1 median 7
NFR8	No external dependency	Needs a server and a local network
NFR9	Met	Met
NFR10	Met	Met

Table 5.14: The two artifacts measured against the requirements.

5.6 When Each Artifact Is the Better Choice

Given these results, **the on-device setup is the safer pick when the network might fail**. NFR8 does not work in small steps. It is all or nothing. It also uses less power on the same headset, and it does not hit the worst stalls seen on the edge setup. People did not show a clear tilt in their own preference. So the argument comes from the data, not from how it felt in use.

The edge design can be the better option too, but only if a local server is there and if you care more about full completion than about the slowest timing. The cover task results point in that direction in both runs. This includes the run where the on-device cover round ended first. That means

the outcome is not just due to which round came earlier. One caveat remains. The cover task changes geometry every three to four seconds. That is slow when compared with a 105 ms round trip. Nothing in these results proves the same edge gain would hold under faster updates or a weaker connection.

Most of all, **the usual reasoning in the literature, which favored edge for lower battery use on the headset, did not match the end result. Over 41 minutes, neither setup throttled. And offloading did not save any energy that could be measured**.

6

Discussion

This chapter lays out what the results suggest. It also compares those findings with earlier studies. Then it explains what the study cannot prove. Finally, it suggests what to try next.

6.1 The Inverted Payload

The goal of the hybrid edge setup is straightforward. Put the heavy work elsewhere, not on the device that cannot dissipate heat. That way, the device should use less power and last longer. From that goal, two outcomes were expected, but **this study found neither to be true.**

The headset never got hot enough to throttle. In 41 minutes of steady meshing, neither design experienced thermal throttling. In the on-device case, the temperature leveled near $49\,^{\circ}\mathrm{C}$ after about ten minutes. After that, it held steady. Heat made and heat lost stayed matched, and the system never reached the point where it would intervene. The issue this approach is supposed to fix never showed up.

Energy went up with offloading. When the real headset differences were accounted for, the edge path used 0.49 W more power in the crossover analysis (Table 5.4). Within a single headset, the gap was between 0.54 and 0.73 W, or roughly 5–7%. The pattern was the same for both units and across all three runs.

Why the measurement design was necessary

That result depends on how the measurement was set up. The two headsets were not identical: one drew 0.79 W more than the other, whatever it was running. **That hardware difference was bigger than the architecture difference being measured.** So a simple side-by-side test gave the wrong answer, because the result depended on which headset ran which architecture. The first run made edge look 13% worse. The second run, with the headsets swapped, made edge look 3% *better*.

The lesson: **when two devices that should be identical differ by as much as the effect being measured, swapping them is not optional. It is the only way to get a reliable answer.**

Why offloading cannot pay for itself here

Three measurements explain why. First, about 95% of the round trip was spent moving data, not processing it. The server needed only about 5 ms of a 94 to 105 ms round trip, so a faster server would have made almost no difference. Second, the downlink carried about 4.5 times more data than the uplink, because the mesh sent back was much larger than the depth frame it was built from. Third, the radio had to carry that data all the time, and that power cost outweighed what was saved by moving the computation off the device.

This is not limited to reconstruction. Offloading works well when the output is smaller than the input. Classification or pose estimation, for example, sends a large image to the server and gets back only a label or a few coordinates. **Dense reconstruction is the reverse: the output is larger than the input, so sending the result back becomes the bottleneck, not the server's computation.** A faster server does not fix that. A useful rule of thumb is to compare the size of the output with the size of the input before offloading a perception task. If the output is larger, the network cost grows with the very data the system is producing.

6.2 The Cost of Depending on Infrastructure

The third run shows this in practice. It was recorded in a hallway chosen to be further from the router. There, the median round trip rose from 94 ms to 651 ms and sustained throughput fell to about a quarter. The edge architecture dropped from about four mesh updates per second to fewer than one, so the mesh used for collisions fell around a second out of date. The

on-device architecture, carried alongside it through the same hallway, was unaffected. The server was not the cause: it took 5.03 ms in the hallway against 4.66 ms in the cluttered room. The slowdown was entirely in the network link.

This changes how NFR8, *independence from infrastructure*, should be read. The requirement treats it as a yes-or-no property, but in practice it is a matter of degree: something as ordinary as moving to another room was enough to change how the system behaved. **A system that works in the living room and fails in the hallway does not need better Wi-Fi. It is a system that breaks without warning.**

For a device meant to be worn all day, that unpredictability may matter more than energy. A steady 5% power penalty is something a designer can plan around. A mesh that quietly falls out of date is simply wrong, and nothing on screen shows it.

6.3 The Interaction That Exposes the Architecture

Free shooting at moving targets received the highest scores for both architectures, with a median score of 7 for accuracy, responsiveness, and stability. This held regardless of where the reconstruction was performed. The only task that showed a clear difference between the two architectures was the cover task.

One explanation is that nothing real moved during the targets task. The targets were virtual and the room stayed as it was scanned, so even an out-of-date mesh was still correct. The task never needed the reconstruction to keep up with a change. **In other words, continuous reconstruction can remain unnoticed by the user until the interaction actually depends on recently changed geometry.** If the study had only measured free-play shooting, it could have incorrectly suggested that users could not distinguish between the two architectures.

This gives a technique that others can reuse: **hide the mesh from the user, move something real, and check whether the interaction still works.** It can be used for any system that claims to keep up with a changing room.

6.4 Perceived Quality and Overall Preference Disagree

Eight of nine participants rated their confidence firing at the moving cover higher under edge, and the ninth rated them equal. Yet when participants were asked at the end to pick one architecture, neither came out ahead:

on four of the five questions, three chose on-device, three chose edge and three chose "both".

So when people answered "which would you choose for a long session", they were judging something other than what they rated round by round. This matters for study design: **a single summary question at the end would have shown no difference between the two architectures, while the per-round ratings separated them on the cover task.**

Why the cover-task result points the way it does

The direction is the surprising part. Edge must send depth and wait for chunks, so a cushion moved into place should appear *later* and shots should pass through it for longer. Participants reported the opposite.

With only the first wave there were two explanations and no way to choose: the edge cover round came first for everyone, so practice, tiredness and heat all favored it or the effect was real. The second wave reversed the order to decide.

The effect did not reverse. With the order flipped, edge was still rated higher by three of four participants and equal by the fourth. Across both waves nobody rated on-device higher. An effect caused by practice would have followed the order when the order changed. This one did not, which rules out presentation order and leaves the architecture itself.

The remaining explanation is that the server, with far more time and power, produces denser and more complete chunks, so shots are blocked more reliably despite arriving later. It finishes in about 5 ms against a 105 ms round trip, so it is under no time pressure. The on-device path must mesh on CPU jobs while the same device renders, which bounds the detail it can afford.

If that reading is right, **offloading trades freshness for completeness**, and for an interaction asking whether a shot is blocked rather than how quickly, completeness won. That is narrower than "offloading feels better": it holds where geometry changes every three to four seconds, which is slow next to a 105 ms round trip. It would not be expected to hold where the world changes as fast as the latency, and the hallway is exactly where it would not. This is the one place in the thesis where edge wins anything, and the win is conditional.

6.5 Relation to Prior Work

AccuMO [14] and Edge-SLAM [3] show that splitting a pipeline beats offloading all of it, for sparse tracking. *This thesis looks at dense TSDF reconstruction, which is far heavier*, and finds that offloading all of integration and meshing did not cause the interaction problems the latency argument predicts, while also failing to deliver the energy saving that motivates it. Their question is still the right one; what this adds is that for dense reconstruction the answer may be that the pipeline should not cross a wireless link at all.

Both of these directions bear directly on the result reported here. The split architectures AccuMO and Edge-SLAM describe keep latency-critical work local and return only refinements, which shrinks the reply rather than the request; the asymmetry measured in Section 5.3.4 is precisely what makes that shape attractive for dense reconstruction. Mesh compression points the same way. This thesis returns the mesh uncompressed, so the downlink reported here is an upper bound rather than a floor, and the energy result should be read as applying to that design. Neither direction was evaluated, and both are taken up in Section 6.8.

Wang et al. [20] argue a break-even point exists between transmission and local computation energy. This study measured it and found that, with the mesh sent back uncompressed, continuous dense reconstruction falls on the wrong side, even over unthrottled local Wi-Fi. A better network would not change that, because the network was already fine: the cost was the radio carrying the mesh back.

Chapter 2 identified a gap in method: this literature leans on simulation and rarely measures perception. This study tried to close it, and the way it partly failed is informative. In the first wave an ordering effect swamped an objectively substantial manipulation, and it took a second wave with an unthrottled control round to get a clean reading.

6.6 Limitations

Sample size. Nine participants. Effect sizes cannot be estimated and no claim about the wider population is made.

Counterbalancing is between waves, not within the design. Order was fixed inside each wave and flipped between them rather than randomized per person. That is enough to show an effect does not follow the order, but it leaves wave and order perfectly correlated. The waves also differ in

other ways such as seven weeks apart, the server defect fixed in between, demographics collected differently. So any difference *between* waves has several possible causes, and none is claimed.

The user-study rooms were not controlled. Nine participants were run across four rooms, and no room identifier was stored in the response files. Room geometry therefore varies between participants and cannot be recovered from the data. Two things bound the consequence. Distance to the access point (which the hallway run identifies as the variable that actually decides whether offloading works) was 1–3 m in every room, matching the cluttered room where the offloaded path performed normally. And because each participant judged both architectures inside one session in one room, room differences cannot generate a within-participant effect, which is where the study's only clear result lives. The exceptions are comparisons *between* participants, and the network degradation block, where each room's baseline link quality sat underneath the induced condition.

The network block is the weakest part. The intended randomization happened in neither wave, so within each one condition and position remain tangled. And all three first-wave conditions capped the downlink below the 27–29 Mbit/s the unthrottled architecture draws, comparing three grades of too-slow rather than spanning adequate to inadequate. The second wave fixed this by adding an unthrottled round, which means the waves cannot be pooled here and four participants carry the block alone.

Who the participants were. Nine people from one social circle, seven of whom had used a headset only "a few times". The group contains almost no experienced users, which matters, because the two most experienced were the only two to report sickness symptoms and one gave the harshest ratings in the study. First-wave demographics were collected from those participants a few weeks after their sessions rather than during them, so they are self-reported but retrospective.

No telemetry during the user study. Nothing anyone said can be checked against what the hardware was doing at the time.

The server defect. Between 17% and 33% of mesh payloads were rejected as malformed, so edge did less work than it should have. Its frame-time figures are therefore better than the truth, and the energy comparison is cautious in its favor: a working edge path would use more power, not less.

Coverage and realism. Two rooms are enough to show that ordinary indoor spaces differ enough to change whether the edge architecture works, but not how often that happens. Rounds ran roughly 1.5 to 7 minutes,

and both heat and sickness build over time, so the study sat in the window where neither has time to appear. A cleared standing area, a hand-moved cushion and short rounds look little like playing at home.

Instrument design. The reverse-scored NASA-TLX Performance item was widely misread in the first wave, and at least one participant answered the network-round items as a measure of how bad the problem was rather than how responsive the system felt.

The cover manipulation was not blinded or scripted. Participants did not know what was being tested, but the researcher did, and the cushion was moved by hand rather than by a machine. It is therefore possible that it was moved slightly differently in one condition than the other, without anyone intending it. Two things keep that small. The same person moved it, to the same instruction, in both rounds of a single session a few minutes apart. And the result goes the opposite way to what the design would predict: the offloaded path has to send depth to the server and wait for the reply, so it should be the slower one to notice a cushion that has just moved. Participants rated it higher instead. Moving the cushion in a way that matched the prediction would have pushed the result the other way.

6.7 Why This Matters

For a developer, the choice of reconstruction architecture is invisible to users for any interaction that does not depend on geometry that just changed. Drop virtual content into a static room and the cheaper architecture wins by default. The architecture only starts to matter when the world changes which is exactly the case scan-and-lock systems cannot handle, and the case this thesis is about.

Theoretically, how good an interaction feels appears to be a property of the task rather than the system: the same system scored at ceiling on one task and mid-scale on another, in one session, for one person. The results also suggest an offloaded task's output-to-input ratio predicts viability better than its computational cost.

Ethically and socially, the edge architecture streams measurements of someone's private space to another machine. This thesis keeps everything in volatile memory and stores nothing, but the wider point stands: **offloading spatial reconstruction turns a privacy question that lived on one device into a network question**, for a kind of data (the shape of somebody's home) that identifies them but is not usually treated as sensitive.

6.8 Future Work

A repeat study could fix most of these limitations: *Give each participant their own random order* of architectures, instead of mirroring the order between the two waves. *Record the headset's measurements during the user sessions*, tagged with each participant. This is the biggest gap in the current study. Run longer rounds, so heat and sickness have time to build up. Randomize the network conditions, and always include one round on a normal network as a baseline. Save the network condition, the round number and the room in the data file itself. And recruit people with a wider range of headset experience, not mostly people with little experience.

Two items from an earlier version of this list were already done between the two waves: the reversed workload question was flipped to match every other scale, and the architecture order was mirrored. Both are reported as changes made during the study, not as the original plan.

There are also three new ideas to explore. First, this study tested only one task, building a 3D mesh of the room. There, the server sent back more data than it received, and offloading did not pay off. Future work could test other tasks, such as recognising objects, to see whether the same rule holds: offloading may pay off when the answer is smaller than the data sent, but not when it is larger. Second, the work could be split between the headset and the server. The headset would quickly build a rough mesh of what is right in front of the user, so that part never falls out of date, and the server would make it more detailed when it can. If the network slowed down, the headset would still have its own rough mesh. AccuMO and Edge-SLAM already split work this way, but only for tracking, not for building a mesh. Third, the mesh sent back could be made smaller, either by compressing it or by sending only the parts of the room that changed. Most of the network traffic was the mesh coming back, so this could reduce the energy cost. It would not fix the hallway problem, however, because that was caused by waiting for replies, not by the amount of data.

6.9 Use of AI Tools

In line with the Department of Computer and Systems Sciences policy on generative AI, the use of such tools is disclosed here.

AI tools (principally Claude, with Gemini also used) were used in four ways. In the literature search they helped identify candidate papers. Every

source found this way was located in its original venue and read by the author before being cited, and no citation appears here that the author has not confirmed exists and says what it is claimed to say. During development they assisted with implementation and debugging of the Unity client and the Metal server, including diagnosing the memory-safety defect in the meshing pipeline. During analysis they were used to write data-processing scripts and to discuss analytical options. For example deriving power from voltage and current rather than battery percentage, and using a crossover design to separate hardware from architectural effects. During writing they supported drafting, restructuring and language editing.

They were not used to generate data, to invent citations, or to produce results. Every figure and statistic in Chapter 5 was computed from the recorded data and checked by the author against the source files. All analytical decisions and conclusions are the author's own, and the author takes full responsibility for the content and originality of this work.

6.10 Conclusion

This thesis built two versions of the same system for continuously rebuilding a 3D model of a room on a Meta Quest 3. One ran entirely on the headset. The other sent the heavy work to a nearby server over Wi-Fi. Both met every functional requirement.

On RQ1, the headset never overheated. Running continuously for 41 minutes, it settled at about $49\,^{\circ}\mathrm{C}$ after ten minutes and stayed there.

On RQ2, offloading used about 6 Mbit/s going up and 27–29 Mbit/s coming back. About 95% of the delay was spent moving data over the network, not computing on the server.

On RQ3, offloading lost on every measurement taken from the devices. It used about 5% more power once the difference between the two headsets was accounted for, it had far worse worst-case frame times, and in the hallway it fell to fewer than one mesh update per second while the on-device version was unaffected. Offloading can only win under narrow conditions: a server must be nearby, the network must be good, and the task must care more about a complete mesh than about smooth frames or battery life.

On RQ4, people could not tell the two versions apart when shooting at virtual targets, where both scored at the top of the scale, and neither version won when people were asked at the end to pick one. They did tell them apart on the cover task, the one task where a real object moved: eight of nine rated edge higher on confidence firing at the moving cover, the ninth

rated them equal, and this held when the order of the rounds was reversed. Workload and sickness showed no difference, and people only noticed a slower network when it was cut hard, to 5 Mbit/s. **So users do not notice how the room is rebuilt until the task depends on something that has just moved.** This shows the direction of the effect, not its size. With nine people, the order changed between waves rather than for each person, and no device measurements were taken during the sessions, so the study shows which way people leaned on one task, but not how large the difference is or whether it holds for users in general.

The contribution is partly a negative result and partly a methodological one. The negative result is that offloading dense reconstruction over Wi-Fi did not pay off on current consumer hardware, at least when the mesh is sent back uncompressed. There are two reasons. First, what came back was bigger than what was sent. Compressing the mesh could narrow that energy gap, which is why it appears in Section 6.8. Second, the system failed without warning in the hallway, most likely because it was further from the router and the signal was weaker. Each reply took much longer to arrive, and because the headset waits for one reply before asking for the next, the mesh fell behind. Compressing the mesh would not remove that weakness: a system that depends on the network still breaks when the network gets bad. This is a useful limit on an approach the literature treats as promising.

The two methodological contributions can be reused by others. The first is the cover task: hide the mesh, move a real object, and check whether the interaction still works. It showed a difference when nothing else did. The second is swapping the devices: when two headsets that should be identical differ by as much as the effect being measured, each one has to run each version, or the result depends on which headset ran what.

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Appendix A — Participant Information and Consent

Participants received a single double-sided sheet containing the information sheet and the consent form. Three versions were used. Version 1.0 went to P01–P03 and version 1.1 to P04–P05; the substantive difference is that v1.1 discloses explicitly that the headset continuously measures room geometry and that, in the offloaded condition, those measurements are transmitted to a laptop on the local network. Version 1.2, dated 25 August 2026, was used for the second wave (P06–P09) and adds two optional items covering an audio-recorded closing conversation and the use of anonymous quotations. Neither option was exercised: no session was recorded. The text reproduced below is version 1.2.

Information provided to participants

Purpose. A Master's thesis at Stockholm University comparing two ways an augmented-reality application on a Meta Quest 3 builds a live three-dimensional model of the surrounding room, so that virtual objects can interact with real furniture and walls. One method performs all computation on the headset; the other sends data to a nearby laptop and receives the result.

Voluntary participation. Participation was entirely voluntary. Participants were told they could stop at any time, without giving a reason and without penalty, and could skip any question. Exclusion criteria were: under 18, pregnancy, epilepsy or a history of seizures, heart conditions, and susceptibility to motion sickness. These criteria were stated on the information sheet and confirmed by the participant in consent item 3; no separate screening instrument was administered.

Procedure. A single session of approximately 45 minutes, comprising a baseline questionnaire, several short rounds of an augmented-reality laser-tag activity, short questionnaires after each round, and a final comparison. Some differences between rounds were not disclosed in advance so that ratings would not be influenced; all were explained in the debrief.

Environmental sensing. Participants were informed that the headset continuously measures the shape of the room — walls, floor, ceiling and furniture — and that in the offloaded condition these depth measurements are transmitted over a private local network to the researcher's laptop in the same room. The data is a grid of distance values describing geometry, not a photographic image. It is processed in volatile memory, is never written to disk on either device, is not transmitted over the internet, and is not retained after the session. No screen or video recording of any session was made.

Risks. Temporary nausea, dizziness, eyestrain, headache or disorientation are known risks of head-mounted displays. Rounds were kept short with breaks, symptom levels were measured throughout, and participants could remove the headset at any moment. The researcher remained present to prevent collisions with physical objects.

Data handling. Questionnaire responses and technical performance logs were collected, identified only by a participant code. The code-to-name list and signed consent forms are stored separately and securely from the data. Questionnaires were completed on the researcher's own device and saved locally, not through any external or cloud questionnaire service, before transfer to access-controlled university storage. Data is retained for one year after completion of the study, after which identifiable data is destroyed. Results are reported in aggregate and participants are not identifiable in any output.

Legal basis. Processing is carried out under the EU General Data Protection Regulation and the Swedish Data Protection Act (2018:218). The legal basis is consent and Stockholm University is the data controller. Participants retain the right to access, rectification and erasure, and may complain to the Swedish Authority for Privacy Protection. Data could be withdrawn up to two weeks after a session, before anonymisation and aggregation.

Consent items

Participants initialled each of the following. Items 6 and 7 were optional.

1. I have read and understood the information above and have had the chance to ask questions.
2. I understand that participation is voluntary; that I may withdraw at any time without reason or penalty, stop and remove the headset at any moment if I feel unwell, and skip any question.
3. I confirm I am 18 or older and that, to the best of my knowledge, none of the exclusion criteria apply to me.
4. I understand that questionnaire responses and technical performance data are collected, and that the headset continuously measures the shape of the room around me; that in one version these depth measurements are sent over a private local network to the researcher's laptop in the same room; and that this room data is used live, not saved, not sent over the internet, and not kept afterward. No photographs, video or screen recording of my session are made.
5. I understand that some differences between rounds are not disclosed in advance so my ratings are not influenced, and will be fully explained in the debrief.
6. *(Optional.)* I consent to the closing conversation being audio-recorded and transcribed.
7. *(Optional.)* I agree to anonymous quotes being used in the thesis and any resulting publications.
8. I understand my data is stored securely under a participant code, kept one year, reported only anonymised and aggregated, and that I may withdraw it up to two weeks after my session.
9. I agree to take part in this study.

Appendix B — Questionnaire Items

All questionnaires were administered through a purpose-built web form on a tablet, completed by the participant between rounds while still seated. Every participant completed ten forms: a baseline, one after each of the four task rounds, one after each of the three network rounds, and two comparison forms at the end. All ninety forms were returned complete.

B.1 Baseline form

Administered once, before the first round.

Virtual Reality Sickness Questionnaire

Nine symptom items [11], each rated *None* (0), *Slight* (1), *Moderate* (2) or *Severe* (3), giving a total from 0 to 27. The same nine items were repeated after each of the four task rounds.

1. General discomfort
2. Fatigue
3. Eyestrain
4. Difficulty focusing
5. Headache
6. Fullness of head
7. Blurred vision
8. Dizziness (with eyes closed)
9. Vertigo

Demographic items

Added for the second wave (P06–P09). The same three questions were put to P01–P05 a few weeks after their sessions, so their answers are self-reported but retrospective.

Question	Options
Which age range are you in?	18–24 | 25–34 | 35–44 | 45–54 | 55+ | Prefer not to say
How do you describe your gender?	Woman | Man | Non-binary | Prefer to self-describe | Prefer not to say
How often have you used a VR or AR headset before today?	Never | A few times | Occasionally | Regularly

Table B.1: Demographic items on the baseline form.

B.2 After each task round

Each of the four task rounds was followed by the VRSQ items above, the NASA-TLX items, and the Likert items for that task.

NASA Task Load Index

Six subscales [7], each on a seven-point scale, applied raw and unweighted.

Subscale	Question	1	7
Mental Demand	How mentally demanding was the task?	Very Low	Very High
Physical Demand	How physically demanding was the task?	Very Low	Very High
Temporal Demand	How hurried or rushed was the pace of the task?	Very Low	Very High
Performance	How successful were you in accomplishing what you were asked to do?	*see below*	*see below*
Effort	How hard did you have to work to accomplish your level of performance?	Very Low	Very High
Frustration	How insecure, discouraged, irritated, stressed and annoyed were you?	Very Low	Very High

Table B.2: NASA-TLX subscales as presented to participants.

The Performance subscale changed direction between waves. In the first wave it was presented in the standard NASA-TLX direction, in which

a low score means good performance — the opposite of every other scale in this study, where 7 always means good. From the second wave onward it was displayed from *Failure* (1) to *Perfect* (7), matching the other items, and reversed back to the standard direction on save so that both waves are stored on one scale. Both the displayed and the stored values are recorded for the second wave and are reproduced in Appendix C. The subscale is excluded from every total reported in Chapter 5.

Custom Likert items

No validated instrument asks whether a live reconstruction was correct about geometry that had just moved, so six items were written for this study. All run from 1 (strongly disagree) to 7 (strongly agree). Item S3 is deliberately repeated across all three contexts so that the same construct can be compared between tasks.

ID	Asked after	Statement
S1	Targets rounds	The reconstructed room geometry looked accurate to the real room.
S3	Targets, cover and network rounds	The system felt responsive to my movement.
S4	Targets rounds	The room reconstruction stayed stable while I moved my head.
S5	Cover rounds	I felt confident shooting at targets behind cover.
R1	Network rounds	The game responded promptly when I fired.
D1	Network rounds	This round felt noticeably different from the earlier rounds.

Table B.3: Custom Likert items, 1 = strongly disagree, 7 = strongly agree.

B.3 After each network round

The three network rounds used items R1, S3 and D1 above, followed by one free-text prompt:

> Were there any real differences you noticed in how the game felt during this round?

Participants saw the rounds labelled only “Test 1” to “Test 3”, so that no severity-suggesting name could anchor their judgment. The mapping from those labels to the induced conditions is in Appendix C.

B.4 Comparison forms

Two forced-choice forms were completed at the end of the session: one comparing the two architectures, one comparing the three network rounds. Both used the same five questions.

ID	Question
C1	Which condition felt most responsive to your movement?
C2	Which condition would you choose for a 30-minute play session?
C3	Which had the best-quality reconstruction of the room?
C4	Which was most comfortable to use?
C5	Which would you most likely recommend to a friend to try?

Table B.4: Forced-choice comparison items.

For the architecture comparison the options were *Edge*, *Standalone* and *Both / no real difference*. The third option exists so that a participant who could not tell the two apart is not forced to name a winner. For the network comparison the options were *Test 1*, *Test 2* and *Test 3*.

Item C2 on both forms was followed by a free-text prompt:

> In one sentence, why did you pick your answer to C2?

All free-text responses are reproduced in Appendix D.

Appendix C — Network Condition Mapping and Excluded Data

C.1 Which "Test" was which condition

Participants saw the three network rounds labelled only "Test 1" to "Test 3", so that no severity-suggesting name could anchor their answers. The mapping from those labels to the induced conditions of Table 3.4 was held in the researcher's run log and is reproduced here; it is not recorded in the response files.

The intended per-participant randomization of this order did not occur in either wave. Four of the five first-wave participants received one sequence and all four second-wave participants received another, which is why condition and position are confounded within each wave. The two sequences run in opposite directions, so an ordering effect visible in one wave can be checked against the other.

Participants	**Test 1**	**Test 2**	**Test 3**
P01, P02, P03, P04	Severe	Moderate	Extreme
P05	Extreme	Severe	Moderate
P06, P07, P08, P09	Extreme	None	Moderate

Table C.1: Mapping from the blinded round labels to the induced network conditions defined in Table 3.4. "None" is an unthrottled round, added for the second wave so that the block spanned adequate to inadequate rather than three grades of inadequate.

The mapping above is cross-checked against the response data: for seven of the nine participants the three rounds carry distinguishable answer triples, and each resolves to exactly one condition. P06 and P09 gave identical answers in two or more rounds and are therefore assigned by their wave's shared sequence rather than by their own data.

C.2 The excluded NASA-TLX Performance subscale

The Performance subscale was asked in two different directions across the two waves and is excluded from every total reported in Chapter 5. Both waves are given here so that the exclusion can be checked.

In the first wave the item was presented in the standard NASA-TLX direction, in which a low score means good performance. The stored value is therefore the participant's raw selection. In the second wave the item was displayed from *Failure* (1) to *Perfect* (7), matching the direction of every other scale in the study, and reversed on save so that both waves are stored on the standard scale. Both the displayed and the stored value are recorded.

		Wave 1 — stored only					**Wave 2**			
Round		**P01**	**P02**	**P03**	**P04**	**P05**	**P06**	**P07**	**P08**	**P09**
Targets, on-device	shown	—	—	—	—	—	3	2	5	7
	stored	3	5	5	5	7	5	6	3	1
Targets, edge	shown	—	—	—	—	—	3	3	7	5
	stored	2	5	6	5	6	5	5	1	3
Cover, on-device	shown	—	—	—	—	—	4	4	1	4
	stored	3	2	4	2	6	4	4	7	4
Cover, edge	shown	—	—	—	—	—	4	5	7	7
	stored	6	6	5	3	7	4	3	1	1

Table C.2: NASA-TLX Performance responses, excluded from all totals. *Shown* is the value the participant selected on screen; *stored* is that value converted to the standard NASA-TLX direction, where 1 is best. The first wave has no separate *shown* row because the item was presented in the standard direction and stored unchanged.

Why the first wave's values are treated as unreliable. Across the two targets rounds the first-wave median is 5 on a scale where 7 means complete failure. The same five participants rated accuracy, responsiveness and stability at a median of 7 in those same rounds, and reported low frustration. Those two readings cannot both be right, and the most economical explanation is that the reversed item was read as though higher meant better. The second wave, which saw the item in the same direction as everything else, produces values that agree with the rest of its own responses.

Appendix D — Free-Text Responses

Two free-text prompts were used. After each network round participants were asked whether they had noticed any real differences in how the game felt during that round. On each of the two comparison forms, item C2 was followed by a request for one sentence explaining the choice.

Fourteen responses were given, by four of the nine participants. All are reproduced below exactly as typed, including spelling and punctuation as entered. Four responses, all from P03, were written in Spanish; the original is given first and an English translation follows in square brackets. Translations are the author's.

D.1 Comparison forms

P	Form	Response
P01	Architecture	It was smooth and responsive; it was fun and engaging to play.
P01	Network	As it was smooth, responsive and comfortable to use
P02	Architecture	more responsive
P02	Network	most responsive
P03	Architecture	I felt it was not delay time
P03	Network	senti que esta opcion era mas preciso y menos tiempo de delay *[I felt that this option was more precise and had less delay.]*

Table D.1: Free-text explanations of item C2 on the two comparison forms.

D.2 Network rounds

The prompt was: “Were there any real differences you noticed in how the game felt during this round?” Round labels are as the participant saw them; the mapping to induced conditions is in Appendix C.

P	Round	Response
P01	Test 1	shooting felt a bit delayed for the cover
P01	Test 2	Shooting the cover was less delayed than the Test 1
P03	Test 1	mi percepción fue que cuando disparaba al objeto dinámico no había ningún delay, lo mismo cuando disparaba a los objetos flotantes. *[My perception was that when I shot at the moving object there was no delay, and the same when I shot at the floating objects.]*
P03	Test 2	en esta oportunidad senti que había un poco de delay, sobre todo cuando disparaba al objeto dinámico. Lo mismo, cuando presionaba el botón para disparar, tuve la percepción de lentitud *[This time I felt there was a little delay, especially when I shot at the moving object. Likewise, when I pressed the button to fire, I had the impression of slowness.]*
P03	Test 3	tuve la percepción, que ha pesar de disparar al objeto de manera adecuada, este recién desaparecía (sensación de acertar el disparo) al segundo intento. *[I had the impression that, despite hitting the object properly, it only disappeared — the sense of the shot registering — on the second attempt.]*
P04	Test 1	Objects were buffering
P04	Test 2	buffering was increased
P04	Test 3	More Buffering

Table D.2: Free-text responses to the network rounds.

D.3 Note on the responses

These responses support the reading given in Chapter 5, that the first wave’s network block could not separate its three conditions.

The two participants who wrote about all three rounds, P03 and P04, both describe the experience getting steadily worse across the block. Their

rounds ran Severe, then Moderate, then Extreme, so the middle round was in fact the mildest of the three. Neither description follows the severity of the conditions; both follow the position of the round.

P04's case is the sharper one, because the free text and the ratings disagree with each other. The text moves from "Objects were buffering" to "buffering was increased" to "More Buffering", while P04's rating on item R1 — whether the game responded promptly — rises across the same three rounds, from 2 to 4 to 7. P03 is at least internally consistent: the text worsens and the R1 rating falls from 7 to 3 to 3.

Only P01 describes a change in the direction the conditions actually moved. P01 wrote of the second round that "Shooting the cover was less delayed than the Test 1", and that round was indeed the milder of the two.

P03 is also the participant with the highest sickness scores, the highest workload totals in three of the four task rounds, and the lowest confidence ratings on the cover task, and was the only participant to describe the reconstruction failing outright rather than merely feeling slow.

Five participants, P05 through P09, left every free-text field blank. The prompt was optional and was not read aloud, so no inference is drawn from the absence of a response.